

The Impact of Artificial Intelligence on Primary School Students' Motivation and Engagement: A Systematic Review

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AI ON MOTIVATION AND ENGAGEMENT

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Jiesi Guo contributes to the conceptualization, methodology, formal analysis, investigation, writing, original draft writing, and review and editing. Ying Ma contributes to the formal analysis, writing, original draft writing, data curation, and review and editing. Hye-Ren Jang contributes to the conceptualization, methodology, formal analysis, and review and editing. Tingting Li contributes to the conceptualization, methodology, formal analysis, and review and editing. Jinran Wu contributes to the methodology, formal analysis, and review and editing. Danling Huang contributes to methodology, data curation, and review and editing. FeiFei Han contributes to the methodology, data curation, and review and editing. Michael Noetel contributes to the conceptualization, methodology, and review and editing. Kewen Liao contributes to the conceptualization, validation, and review and editing. Xin Tang contributes to the conceptualization, validation, and review and editing. Kui Xie contributes to the conceptualization, methodology, and review and editing.

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AI ON MOTIVATION AND ENGAGEMENT

Abstract

The application of artificial intelligence (AI) in primary education has attracted growing attention, with increasing studies exploring its potential to enhance learning. However, prior reviews often overlook primary school students and conceptualize motivation and engagement as single-dimensional. This systematic review, guided by Self-Determination Theory and the multidimensional engagement framework, analyzed 33 empirical studies to evaluate AI's impact on primary students' motivation and engagement. Findings show that AI can fulfil students' basic psychological needs for autonomy, competence, and relatedness, and effectively promote general, intrinsic, and extrinsic motivation (particularly external and identified regulation) among primary school students. AI also enhances engagement across behavioral, cognitive, emotional, and agentic dimensions. However, when AI responses are complex, mechanical, or less engaging than human-mediated tasks, it may lead to negative experiences including frustration, boredom, or reduced motivation. AI's impact varies across contexts by satisfying the need for competence and fostering intrinsic motivation and emotional engagement in STEM subjects, while promoting relatedness in humanities and social sciences. Younger students generally show greater motivation and engagement with AI-supported learning than older ones. Different types of AI tools—personalized, emotionally supportive, interactive, immersive, and feedback-oriented AI—offer distinct advantages in fostering motivation and engagement. Compared to traditional AI, generative AI uniquely enhances learning value perception and strategy adjustment. This review provides theoretical and empirical support for integrating AI in primary education and highlights the need to prioritize motivation and engagement in AI-supported instructional design.

Keywords: Artificial intelligence (AI); Primary school students; Motivation; Engagement; Self-determination theory (SDT)

1. Introduction

With the rapid advancement of technology, artificial intelligence (AI) has become a transformative force across various domains (Zafari et al., 2022). It seems capable of simulating human cognitive abilities such as knowledge acquisition, reasoning, and adaptive learning, so AI is increasingly becoming an integral component of educational practice (Alasgarova & Rzayev, 2024). Research spanning multiple educational contexts, from primary to higher education, indicates that AI enhances student interaction and personalized guidance, thereby enriching the learning process (Yuan & Liu, 2025).

Specifically, in primary education, AI exhibits significant potential for transforming learning experiences (Aravantinos et al., 2024). A variety of AI-driven tools have been developed, including chatbots, automated grading systems, virtual teaching assistants, and intelligent tutoring platforms. These tools support diverse forms of interaction aimed at facilitating student learning (Pardamean et al., 2022; Song et al., 2023; Yim & Su, 2025). For instance, AI can analyze individual students' learning data to offer real-time feedback, assisting primary school students in developing a clearer understanding of their learning progress (Chiu et al., 2024). Additionally, AI-generated adaptive learning content can dynamically adjust students' individual paces and preferences, thus allowing personalized and flexible learning experiences (Sajja et al., 2024). However, alongside these advantages, AI integration in education also introduces challenges (Fuchs, 2023). An overreliance on AI learning tools, especially generative AI offering immediate answers, may reduce students' motivation and engagement for exploration by bypassing independent thinking (Suyu et al., 2024).

As AI continues to permeate educational settings, it becomes essential to explore its impact on critical aspects of learning experiences: motivation and engagement (Fredricks et al., 2019; Guo et al., 2024; Ryan et al., 2023). These constructs are closely related yet theoretically distinct and complementary (Skinner & Raine, 2022). While motivation is defined as the internal driving force that initiates, sustains, and directs students' learning behaviors intentionally and purposefully (Ryan & Vansteenkiste, 2023), engagement refers to the extent to which students actively and productively invest in academic activities or tasks (Jang et al., 2024). In other words, motivation is the underlying sources and intentions behind learning,

AI ON MOTIVATION AND ENGAGEMENT

whereas engagement is the external expressions of motivation through students' actions and participation (Martin et al., 2017; Xie et al., 2024). Considering both constructs simultaneously provides a more holistic understanding of AI's impact on students, capturing internal drivers and external involvement (Skinner & Raine, 2022).

In primary education, learning motivation and engagement are particularly crucial to students' academic development (Li et al., 2024). Compared to later educational stages, students in primary school have not yet developed stable academic self-concepts and learning strategies, leaving greater room for motivational interventions. Primary school represents a crucial window for shaping learning motivation. At this stage, cognitive readiness is still forming, and with appropriate guidance, there is significant promise to foster lasting motivation and engagement (Cantor et al., 2019; Guo et al., 2024; Jeon, 2024). Thus, primary school students represent a critical group for examining AI's potential to enhance motivation and engagement in learning. Moreover, as the cognitive abilities of primary school students are still developing, they rely heavily on external scaffolding such as teacher guidance, learning resources, and technological tools. Compared to secondary and higher-level students, primary school children are less capable of independently regulating their learning. They are more sensitive and responsive to AI features like prompts, feedback, and guidance, making their motivation and engagement more susceptible to AI (Gilmore, 2018). Consequently, focusing on this age group allows for a clearer understanding of AI's practical value in contexts where external support is essential (Dong, Choi, et al., 2024).

Despite the growing prevalence of AI applications in primary education, a comprehensive understanding of their specific effects on students' motivation and engagement remains limited (Aravantinos et al., 2024). Therefore, this paper aims to conduct a systematic review to address AI's role in fostering motivation and engagement among primary school students by synthesizing empirical studies and providing a holistic perspective. The outcomes of this review will contribute valuable theoretical insights and practical guidance for the effective implementation of AI in primary educational contexts (Jeon, 2024).

2. Frameworks for the Current Review

AI ON MOTIVATION AND ENGAGEMENT

To guide this systematic review, the frameworks of self-determination theory (SDT) and multidimensional engagement are adopted, which provide the analytical lens for identifying, analyzing, and interpreting studies related to primary school students' motivation and engagement in the context of AI-supported learning.

2.1 Self-determination Theory (SDT) Framework

Self-determination theory (SDT) is a motivation theory rooted in basic psychological needs, involving individuals experiencing a sense of ownership in both commencing and regulating their actions, intimate connections with others, and effectiveness in given tasks (Ryan & Deci, 2017). Compared to other motivational theories, such as expectancy-value theory and achievement goal theory, which primarily focus on cognitive processes—SDT integrates mechanisms depending on the extent of basic psychological needs satisfaction. This allows for a more comprehensive understanding of motivation by addressing both cognitive (judgement-based) and need (desire-based) components (Ryan & Deci, 2020). Its broad applicability and explanatory power in educational contexts have been widely demonstrated (Bandhu et al., 2024; Galindo-Domínguez et al., 2025; Howard et al., 2021). Therefore, rather than constraining our analysis to only motivational constructs explicitly defined within SDT, we utilize SDT as a comprehensive organizational framework to systematically categorize and interpret the diverse motivational phenomena observed across studies. This approach allows us to integrate findings on various motivational constructs while maintaining theoretical coherence through SDT's established conceptual structure.

According to SDT, motivation becomes more self-determined when individuals' basic psychological needs are satisfied (Ryan & Deci, 2017). Psychological needs consist of three fundamental components: autonomy, competence, and relatedness (Deci & Ryan, 1985; Ryan & Deci, 2020). Autonomy refers to the need for personal endorsement in the behaviour they initiate and regulate. Competence represents the need to be effective and capable in given tasks. Relatedness refers to the need to establish meaningful social connections with others, such as experiencing a sense of belonging or mutual warm caring (Ryan & Deci, 2017; Ryan & Vansteenkiste, 2023). Fulfilling these three psychological needs significantly facilitates a more

AI ON MOTIVATION AND ENGAGEMENT

internalized and sustained intention to engage in learning activities. (Ryan & Deci, 2020; Ryan et al., 2021).

SDT conceptualizes motivation as a continuum that can be broadly categorized into intrinsic motivation, extrinsic motivation, and amotivation as shown in Figure 1 (Ryan & Deci, 2020). Intrinsic motivation refers to an individual's voluntary engagement in an activity or task driven by the pursuit of inner joy and interest without external inducements such as rewards or punishments (Eager & Brunton, 2023; Ryan et al., 2023).

Extrinsic motivation involves engaging in a behavior or task for instrumental purposes (Ryan & Vansteenkiste, 2023). Although extrinsic motivation is non-autonomous and non-self-determined (Ngo et al., 2017), SDT argues that extrinsic motivation can also exhibit varying degrees of self-determination as individuals internalize external values and norms (Ryan & Deci, 2000). Depending on the internalization level determined by the extent of needs satisfaction, Ryan and Deci (2017) categorized extrinsic motivation into four subtypes: external regulation, introjected regulation, identified regulation, and integrated regulation. External regulation refers to behavior entirely driven by external factors, such as rewards or punishments, without any internalization (Ryan & Deci, 2020). Introjected regulation describes partially internalized pressures, such as performing tasks to avoid guilt or gain social approval (Ryan & Vansteenkiste, 2023). Identified regulation occurs when individuals recognize the inherent value and importance of a behavior or task and engage willingly due to personal significance rather than external pressures (Ryan & Deci, 2020). Integrated regulation represents the highest internalization level, where external values are fully assimilated into an individual's self-concept, guiding voluntary and stable self-regulation aligned with personal values (Ryan & Vansteenkiste, 2023).

Lastly, amotivation represents the low end of the motivational continuum, reflecting a complete lack of intention or self-determined regulation (Ryan & Deci, 2017). This passive state, marked by the absence of intrinsic motivation, typically arises from either a perceived lack of value or interest or a belief that the outcome is unattainable.

2.2 Multidimensional Engagement Framework

AI ON MOTIVATION AND ENGAGEMENT

Engagement refers to the extent to which students actively and productively invest in academic activities or tasks (Jang et al., 2024; Reeve et al., 2025a). Engagement is often conceptualized as a multidimensional construct with four dimensions: behavioral engagement, cognitive engagement, emotional engagement, and agentic engagement (Michou et al., 2023; Reeve et al., 2020). Behavioral engagement refers to the observable actions individuals take to achieve learning tasks or goals (Reeve et al., 2020). It includes effort, persistence in challenges, and consistent practice (Reeve et al., 2025a). Cognitive engagement involves the mental effort in learning, including focus and attention control, conceptual reasoning, problem-solving, and critical thinking (Jang et al., 2024). Emotional engagement reflects affective responses—both positive and negative—toward learning experiences, including feelings like enthusiasm, joy, boredom, frustration, and anxiety (Michou et al., 2023). Agentic engagement refers to proactive, constructive actions students make to enrich their learning experiences (Reeve et al., 2022; Reeve & Jang, 2022). This includes actively asking questions, expressing preferences or needs, seeking help or feedback, and making suggestions (Jang et al., 2024).

Having established these theoretical frameworks to guide our analysis, we now turn to examining how existing systematic reviews have addressed AI's impact on motivation and engagement, identifying important gaps in the literature that the current study aims to address.

3. Existing Systematic Reviews on AI in Motivation and Engagement

Reviews on AI in education have touched upon its impact on learning motivation and engagement, and have highlighted that personalized content, synchronous interactions, and immediate guidance can enhance motivation (Deng et al., 2025; Labadze et al., 2023; Wollny et al., 2021).

Several reviews have specifically focused on the impact of AI on learning motivation and engagement (Artemova, 2024; Javaid, 2024). For instance, Al Nabhani et al. (2024) conducted a bibliometric analysis with a systematic review of 12 studies (2013-2023), highlighting AI technologies' potential to enhance student motivation. (Al Nabhani et al., 2024). Similarly, Javaid (2024) focused on AI applications in language education across various age groups,

AI ON MOTIVATION AND ENGAGEMENT

demonstrating how AI tools can foster student engagement and interest in learning (Javaid, 2024). Artemova (2024) reviewed 69 articles indexed in Scopus from 2020, examining AI in education from the perspective of activity theory. Their review emphasized the interactive and adaptive nature of AI tools in support of students' extrinsic motivation; it noted an evident lack of research into AI's impact on intrinsic motivation (Artemova, 2024).

While providing valuable initial insights into the relationship between AI and student motivation and engagement, these reviews exhibit significant limitations of research scope and analytical depth. First, previous reviews typically present overly generalized findings without adequately differentiating between educational contexts or age groups, particularly neglecting primary school students (Javaid, 2024). Although some reviews have mentioned young learners, they often fail to address the unique characteristics of primary school. Primary students at this stage are more reliant on intuitive, engaging, and interactive instructional approaches, making AI tools potentially valuable for enhancing motivation and participation. Additionally, in educational systems where primary teachers are required to teach all subjects without specialized backgrounds (e.g., the United States), students may particularly benefit from AI-assisted learning. However, current literature provides insufficient evidence on whether and how AI effectively supports student motivation and engagement specifically in primary education settings.

Furthermore, previous reviews often treated motivation and engagement as unidimensional constructs, evaluating AI's impact broadly without accounting for their internal complexity of these psychological processes. Motivation is often assessed as a singular construct rather than distinguishing among various motivational subtypes, such as intrinsic motivation, identified regulation, or introjected regulation (Artemova, 2024). Similarly, engagement is rarely differentiated by its distinct dimensions: behavioral, emotional, cognitive, and agentic engagement, even though each engagement serves specialized functioning in learning. (Al Nabhani et al., 2024; Artemova, 2024). Recognizing the complexity of motivation and engagement is essential because different AI tools may impact specific aspects of motivation and engagement differently. For instance, adaptive feedback systems might primarily enhance cognitive engagement and autonomous forms of motivation, while

AI ON MOTIVATION AND ENGAGEMENT

emotionally supportive AI agents might better foster emotional engagement and promote relatedness-driven motivation. Without such distinctions, evaluations of AI's educational impact remain superficial and overly general and fail to capture its specific mechanisms and effects.

Therefore, this systematic review aims to address these critical gaps by investigating the impact of AI applications on students' motivation and engagement, specifically in primary education. Guided by SDT and the multidimensional engagement framework outlined earlier, this review systematically analyzes empirical studies exploring the specific effects of AI technologies on different types of motivation and multidimensional engagement. By providing a more detailed and nuanced classification and analysis, this review seeks to deepen theoretical understanding and offer practical guidance for effectively integrating AI into primary education settings. Furthermore, it aims to inform innovative teaching practices and support educators in better navigating the rapid transformations shaped by AI advancements.

Specifically, the research questions guiding this systematic review are:

RQ1: Impact of AI on motivation: What is the impact of AI tools and resources on primary school students' learning motivation? Which aspects of motivation (e.g., intrinsic/extrinsic motivation, psychological needs) are influenced by AI tools?

RQ2: Impact of AI on engagement: What is the impact of AI on primary school students' learning engagement? Which aspects of engagement (e.g., behavioral engagement, cognitive engagement, emotional engagement, agentic engagement) are influenced by AI tools?

RQ3: Types of AI tools affecting motivation and engagement: Which types and specific design features/principles of AI tools and resources affect primary school students' motivation and engagement?

RQ4: Differential Effects of AI: How do the effects of AI tools and resources vary among primary school students based on factors such as grade level, gender, subject area, and generative capabilities?

4. Methodology

AI ON MOTIVATION AND ENGAGEMENT

4.1 Protocol and Registration

This systematic review follows the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, which consists of four key stages (Liberati et al., 2009). The four stages include identification, screening, review, and inclusion (Liberati et al., 2009). The research protocol was registered on December 20, 2024, in the Open Science Framework (OSF) under the title: “*The Impact of Artificial Intelligence on Primary School Students’ Motivation and Engagement: A Systematic Review.*” (https://osf.io/sbtf2/?view_only=4205cac53e93423cb46d8a7a056beebebc).

4.2 Identification

The initial search for this systematic review was conducted on December 20, 2024, with subsequent searches continuing until February 12, 2025, to ensure the inclusion of the most recent literature. A systematic search was performed across several electronic databases, including Web of Science, Scopus, IEEE Xplore, Taylor & Francis, Wiley, ERIC, ACM, PsycINFO, and Education Source (EBSCO). These databases were selected as they are recognized as major academic publishing sources, ensuring the quality and reliability of the systematic review. Moreover, these databases comprehensively cover key journals in education, psychology, and technology. To mitigate potential publication bias, we intentionally expanded our search beyond peer-reviewed journals to include preprint articles from the arXiv platform, conference proceedings, dissertations, and technical reports. This comprehensive approach helped capture unpublished or in-progress research that might not have reached traditional publication channels, particularly important in rapidly evolving fields like educational technology and AI, where significant work may exist outside formal journal publications. Additionally, we conducted backward and forward citation searching on included articles to identify any relevant studies that may have been missed in the database searches, further reducing the risk of publication bias.

A structured search strategy was employed in this systematic review. The search terms were derived from previous systematic reviews and relevant literature related to the research

AI ON MOTIVATION AND ENGAGEMENT

questions. All search terms were applied to titles and abstracts to ensure the retrieval of relevant studies. Within each category, search terms were combined using Boolean operators: an asterisk (*) was used to account for variations in word endings, while terms within the same group were connected using “OR”. Different groups of search terms were then linked using “AND” to refine the search results. The final search terms used in this study can be found in Table S1 in the supplemental materials.

4.3 Screening and Review

A total of 8,997 records were generated based on the search strategy. The retrieved records were imported into EndNote and subsequently transferred to Covidence for record management. After automatic and manual deduplication, 1,992 articles were removed, leaving 7,005 imported articles for screening. Two independent reviewers conducted an initial screening based on titles and abstracts, followed by a full-text screening of the articles that met the inclusion criteria. Both screenings were performed independently. Table 1 presents the inclusion and exclusion criteria applied in this review. Corrections were needed in just 2% of cases, and any inconsistencies at each screening stage were resolved through discussion between the two reviewers.

According to the inclusion and exclusion criteria, 6,801 articles were excluded based on title and abstract screening. In the full-text screening stage, an additional 173 articles were removed for not meeting the inclusion criteria, as illustrated in the PRISMA 2020 flow diagram showing the detailed study selection process (see Figure 2). Additionally, one article was removed due to content duplication with an included study despite having a different title. Ultimately, 31 articles were included for backward and forward citation searching in the next step, using the database Scopus. As a result of these supplementary searches, two additional studies meeting the inclusion criteria were incorporated into the review. 33 articles were included in the final analysis.

4.4 Quality Assessment

The 33 included studies were independently assessed by two reviewers using the Joanna

AI ON MOTIVATION AND ENGAGEMENT

Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental Studies and the JBI Critical Appraisal Checklist for Qualitative Research. As the JBI tools do not provide a unified tool for mixed-methods studies, the quantitative and qualitative components were evaluated separately. The mean quality score was 5.6/9 (SD =) for quantitative studies and 6.7/10 (SD =) for qualitative studies. Lower scores in quantitative studies were mainly due to reliance on observational designs or a predominant focus on qualitative inquiry. Importantly, the quality appraisal was used to summarize the evidence base, and no studies were excluded based on quality scores. Full assessment details are provided in Table S2 and Table S3 in the supplemental materials.

4.5 Qualitative Synthesis

The data analysis in this review was conducted using thematic and content analysis (Cohen et al., 2018). Thematic analysis was employed to identify and categorize key themes from the selected studies, which were then mapped onto a content analysis coding framework to examine data distribution across pre-defined coding categories (Cohen et al., 2018). Guided by the conceptual frameworks of motivation and engagement adopted in this review, the coding scheme for content analysis included the following categories: Intrinsic Motivation, Identified Regulation, Introjected Regulation, External Regulation, Autonomy, Competence, Relatedness, Behavioral Engagement, Cognitive Engagement, Emotional Engagement, and Agentic Engagement. General Motivation and General Engagement were additionally used to code findings without clearly specified aspects. All authors reviewed and approved the coding framework. Data extraction and coding for the 33 included studies were independently performed by one reviewer. To establish coding reliability, a randomly selected 10 studies were independently coded by a second reviewer. The inter-rater agreement between the two coders was 89%, indicating a high level of consistency (McHugh, 2012). Any discrepancies in coding were discussed and resolved to reach a consensus.

Additionally, key information from the 33 included studies was extracted into a customized Microsoft Excel systematic review table. Extracted data included publication details (including first author, affiliated country, article title, publication year, and journal title

AI ON MOTIVATION AND ENGAGEMENT

or conference name) and study characteristics (including number of participants, age, gender, and country). Table 2 presents the extracted information.

5. Results

There is a growing interest in research on AI in primary school students' motivation and achievement. There have been at most 4 papers per year since 2020 (see Figure 3). Since 2023, however, there has been a noticeable surge in publications, with the number rising to 6 in 2023 and more than doubling to 13 in 2024. There has also been an increasing diversity in the number of tools being used, with generative AI emerging and becoming the most frequently used in 2024. Asia had the highest number of studies ($n = 20$; 60.61%), North America ranked second ($n = 7$; 21.21%), followed by Europe ($n = 3$; 9.10%) and Africa ($n = 2$; 6.06%) (see more details in Figure S1). Regarding subject areas, the studies can be categorized into STEM disciplines and Human and Social Disciplines, with a greater emphasis on STEM disciplines ($n = 18$), while Human and Social Disciplines ($n = 14$) received comparatively less attention, as shown in Figure 4. Within STEM Disciplines, research was primarily focused on science ($n = 8$), making it the most studied subject. This was followed by mathematics ($n = 5$) and computer science and artificial intelligence ($n = 3$), along with STEM education ($n = 2$), reflecting the growing importance of AI and interdisciplinary integration in education research. Studies in human and social disciplines ($n = 14$) mainly concentrated on language and literacy ($n = 12$), with social sciences ($n = 2$) also represented, highlighting AI's extensive applications in language learning and literacy education.

Regarding research methods, 2 studies utilized qualitative designs, 15 quantitative designs, and 16 mixed-methods approaches. Among studies that employed quantitative methods, 9 were pre-experimental (only pre/post-test without control group), 7 were quasi-experimental (4 post-test and control, 3 pre/post-test and control), 6 were randomized controlled trials (RCTs), and 7 were observational designs.

5.1 RQ1: Impact of AI on Motivation

5.1. *Quantitative Findings on Psychological Needs*

AI ON MOTIVATION AND ENGAGEMENT

According to SDT, quantitative studies suggest that AI influences primary school students in three key areas: Autonomy ($n = 1$), Competence ($n = 8$), and Relatedness ($n = 6$), as shown in Table 3. Although Moltudal et al. (2020) found that the effects of AI on these aspects were not statistically significant before and after AI tool usage, multiple studies consistently indicated improvements in Competence, particularly concerning self-efficacy. Six studies reported that AI helped enhance students' self-efficacy among students using AI technologies compared to non-AI users or their baseline self-efficacy. However, one study found a non-significant change in students' confidence (Lee et al., 2023). For Relatedness, studies identified AI's positive influence across multiple dimensions, which included students' interactions with AI learning tools, such as social interaction ($n = 3$), communication experience ($n = 1$), and enhanced peer collaboration ($n = 1$). Collectively, these findings demonstrated AI's potential in fostering students' interpersonal connections and teamwork.

5.1.2 Qualitative Findings on Psychological Needs

Qualitative research similarly demonstrated AI's impact on the three aspects of SDT: Relatedness ($n = 15$), Autonomy ($n = 2$), and Competence ($n = 4$), as shown in Table 4 (see more details in Table S4 in supplemental materials). For Autonomy, students valued AI's capability to facilitate self-paced learning ($n = 1$) and independent learning ($n = 1$). Feng and Wang (2023) and Jeon (2024) reported that students perceived AI as providing opportunities for autonomous learning, allowing them to complete tasks independently at their own pace. For Competence, all four studies indicated that AI increased students' confidence in learning, such as mathematics (Lee, 2024; Zheng & Tse, 2023) and English (Huang et al., 2008). Relatedness emerged as the most frequently discussed aspect, with findings highlighting two distinct types of connections. One aspect was that AI enhanced human-human connections by improving team cooperation ($n = 4$), as reflected in students' reports of group role-plays and idea-sharing fostering team spirit and collaboration (Feng & Wang, 2023; Tai & Chen, 2024). Another aspect involved students' relationality with AI, encompassing their social interaction ($n = 3$) and comfortable communication experiences ($n = 5$), where they described interactions with AI as social connections resembling friendships (Huang et al., 2008) and reported realistic (Jeon,

AI ON MOTIVATION AND ENGAGEMENT

2024) and comfortable (Dong, Mitchell, et al., 2024) communication experiences. Overall, qualitative studies indicate AI's positive contributions to fulfilling students' psychological needs, particularly relatedness, thereby supporting their motivation and engagement in learning activities.

In summary, 15 quantitative findings under the three basic psychological needs were identified, with 11 showing positive effects of AI on students' autonomy, competence, and relatedness, and the remaining four reporting no statistically significant changes.

Additionally, 19 qualitative findings were identified, all of which demonstrated positive effects, suggesting that AI shows strong potential in supporting students' basic psychological needs.

5.1.3 Quantitative Findings on Motivation

Quantitative data indicate that AI's impact on primary school students' motivation is reflected in General Motivation (n = 4), Intrinsic Motivation (n = 19), Identified Regulation (n = 4), External Regulation (n = 2), and Amotivation (n = 1). Findings related to General Motivation were inconsistent, as shown in Table 5.

Specifically, while one study found that AI significantly enhanced students' motivation (Cheng et al., 2024), another reported no significant change (Chen et al., 2024). Additionally, two studies using descriptive statistics indicated that students showed low recognition of AI's role in improving motivation (Wilson et al., 2021; Wilson et al., 2024).

For Intrinsic Motivation, AI primarily influenced three subdimensions: interest (n = 12), curiosity (n = 2), and future intention (n = 4), suggesting a positive relationship between AI tool usage and intrinsic motivation compared to traditional methods or baseline states (Chettaoui et al., 2020; Feng & Wang, 2023; Wang et al., 2021). However, some findings indicated non-significant or negative impacts. Chen et al. (2024) found a slight, non-significant increase in interest, while Dong, Choi, et al. (2024) and Wilson et al. (2024) reported that AI usage led to a decline in curiosity and lower agreement regarding future intention.

In terms of Identified Regulation, research findings suggest that AI's positive impact is reflected in students' perceived usefulness of AI (n = 2). However, one study found no

AI ON MOTIVATION AND ENGAGEMENT

significant change in students' value perception of learning content (Lee et al., 2023). As for External Regulation, Chen et al. (2024) observed a significant increase in External Regulation among students exposed to AI tools compared to those in the control group. Conversely, Amotivation significantly increased in one study after students engaged with AI-based learning (Moltudal et al., 2020)

5.1.4 Qualitative Findings on Motivation

Qualitative findings revealed similar motivational dimensions affected by AI: General Motivation (n = 5), Intrinsic Motivation (n = 20), Identified Regulation (n = 4), External Regulation (n = 2), and Amotivation (n = 1), as shown in Table 6 (see more details in Table S5 in supplemental materials). Intrinsic Motivation was the most frequently studied dimension, specifically in interest (n = 6), curiosity (n = 1), and future intention (n = 4). Students consistently expressed enhanced interest across diverse subjects, including STEAM (Dong, Choi, et al., 2024), mathematics (Zheng & Tse, 2023), poetry (Chen et al., 2024), and reading (Lee, 2024). For Future Intention, three studies found that students indicated an increased willingness to engage more with AI-supported learning tasks (Aslan et al., 2024; Dong, Choi, et al., 2024; Huang et al., 2008), with some explicitly stating their readiness to dedicate extra time and effort in learning activities (Zheng & Tse, 2023).

For Identified Regulation (n = 3), studies indicated that AI improved students' attitudes toward AI tools and learning content, primarily through improved perceptions of AI usefulness (n = 2) (Jeon, 2024; Lee, 2024) and learning content value (n = 1) (Zheng & Tse, 2023). Regarding External Regulation, one study reported that AI tools incorporating rewards, such as applause and game-like interactions, effectively enhance External Regulation (Burak & Gultekin, 2024). However, some qualitative studies revealed negative motivational impacts. General Motivation (n = 1) and Amotivation (n = 1), one study in each category suggested that AI might, in certain situations, reduce student motivation. Jeon (2024) found that the perceived lack of appeal in AI diminished students' general motivation to perform language tasks with chatbots. Similarly, in Moltudal et al. (2020)'s study, students experiencing challenges in collaborating with AI developed increased feelings of helplessness, leading to greater

AI ON MOTIVATION AND ENGAGEMENT

Amotivation.

In summary, a total of 31 quantitative findings on motivation were identified, with 21 reporting positive effects of AI on primary students' motivation across four dimensions: General Motivation, Intrinsic Motivation, Identified Regulation, and External Regulation. The remaining 10 indicated either negative effects or no significant change. Among 33 qualitative findings, most showed positive effects, with only one reporting a negative impact on General Motivation. Notably, one study found increased levels of Amotivation across both quantitative and qualitative data, a finding that indicates a negative impact of AI, as Amotivation represents the absence of motivation.

5.2 RQ2: Impact of AI on Engagement

5.2.1 Quantitative Findings on Engagement

Quantitative research identified AI's impact on primary school students across General Engagement (n = 6), Behavioral Engagement (n = 5), Cognitive Engagement (n = 5), Emotional Engagement (n = 21), as shown in Table 7. For General Engagement, five studies reported significantly increased active student participation in AI-assisted learning environments, while one indicated a positive but non-significant effect (Zhexenova et al., 2020). However, these studies did not differentiate engagement into specific dimensions.

For specific dimensions, AI was found to support Behavioral Engagement, such as increased learning time (n = 2), learning volume (n = 2), as well as balancing learning content (n = 1). For example, Feng and Wang (2023)'s study observed, students in AI-assisted bilingual reading environments exhibited more balanced participation across English and Chinese content than those using traditional printed materials. For Cognitive Engagement, AI was shown to enhance higher-order thinking (n = 3) and sustained concentration (n = 2). Emotional Engagement emerged as the most extensively examined dimension, revealing both predominantly positive emotions, including enjoyment (n = 12), satisfaction (n = 4), and excitement (n = 1). Enjoyment was the most consistently reported benefit, with studies showing higher enjoyment ratings among students using AI compared to those using other

AI ON MOTIVATION AND ENGAGEMENT

learning methods and increased positive emotional responses pre- and post-AI implementation. However, negative emotional effects were also documented, such as frustration ($n = 1$). As reported in Tai and Chen (2024)'s study, a small proportion of students experienced frustration due to AI's advanced vocabulary and lengthy responses, regardless of whether they used the robot individually (25%) or in paired interactions (17.86%).

5.2.2 Qualitative Findings on Engagement

Qualitative studies provided additional insights into AI's impact on primary school students' engagement, addressing dimensions of General Engagement ($n = 3$), Behavioral Engagement ($n = 3$), Cognitive Engagement ($n = 5$), Emotional Engagement ($n = 20$), and Agentic Engagement ($n = 5$), as shown in Table 8 (see more details in Table S6 in supplemental materials). Consistent with quantitative findings, Emotional Engagement remained the most frequently studied dimension, covering enjoyment ($n = 8$), satisfaction ($n = 4$), excitement ($n = 4$), frustration ($n = 2$), and boredom ($n = 2$). Students frequently expressed enjoyment and excitement in interactions with AI; however, negative emotional responses such as frustration and boredom were reported when AI tasks became repetitive or interactions felt overly mechanical (Jeon, 2024; Moltudal et al., 2020). For Cognitive Engagement ($n = 5$), studies highlighted AI's positive effects on students' higher-order thinking. Additionally, one study found that children demonstrated patience and attention while waiting for the robot to initiate actions, reflecting deeper cognitive engagement (Dong, Choi, et al., 2024). Regarding Behavioral Engagement ($n = 3$), qualitative studies reinforced quantitative findings, demonstrating that AI positively influenced learning time and learning volume, as evidenced by students continuing AI interactions during free playtime. Furthermore, Cheng et al. (2024)'s study suggested that AI also positively affected participation, actively re-engaging students who previously showed minimal engagement (e.g., repeatedly responding "Hi" or "I don't know"). Furthermore, qualitative studies found that AI also enhanced primary school students' Agentic Engagement, reflected in tool application ($n = 3$), question asking ($n = 1$), and strategy adjustment ($n = 1$). Students actively utilized additional resources to enhance their interactions with AI tools (Burak & Gultekin, 2024; Jeon, 2024; Tai & Chen, 2024). Additionally, students

AI ON MOTIVATION AND ENGAGEMENT

exhibited proactive inquiry and reflective strategy adjustments in their interactions with AI, asking higher-order questions (Dong, Choi, et al., 2024; Tai & Chen, 2024). These findings highlight how AI not only encourages passive involvement but actively promotes students' agency in their learning processes.

In summary, among the 37 quantitative findings on engagement, 35 reported positive effects of AI on general, behavioral, cognitive, and emotional dimensions. One study found no significant change in general engagement, while another reported increased frustration within emotional engagement—reflecting a negative emotional response. Of the 36 qualitative findings, most also indicated positive effects across these four dimensions, with additional evidence for agentic engagement, suggesting that AI may foster students' active participation. However, four qualitative studies noted increases in frustration and boredom, highlighting potential negative emotional experiences during AI-supported learning.

5.3 RQ3: Types of AI Tools Affecting Motivation and Engagement

Based on the types of AI tools adopted, as shown in Figure 3, those that impact motivation and engagement can be classified into five distinct categories: Personalized and Adaptive AI, Emotionally Supportive AI, Collaborative and Interactive AI, Immersive and Multimodal AI, and Feedback-oriented AI.

5.3.1 Personalized and Adaptive AI

Personalized and adaptive AI, including adaptive learning systems (Burak & Gultekin, 2024; Lee, 2024; Moltudal et al., 2020; Zohra Lhafra & Abdoun, 2023), intelligent recommendation systems (Chang & Chen, 2022; Wang et al., 2022), and intelligent tutoring systems (Ward et al., 2013), tailors learning activities and content to student's individual needs, interests, and knowledge levels. By analyzing students' performance data, these AI tools dynamically adjust the difficulty and complexity of content to align with their abilities (Qian et al., 2023). For example, Burak and Gultekin (2024) demonstrated that adaptive storylines and personalized characters enhanced student engagement by aligning closely with individual preferences. Similarly, Ward et al. (2013) reported an intelligent tutoring system that adjusted

AI ON MOTIVATION AND ENGAGEMENT

instructional content according to children's mastery levels in mathematics, offering in-depth discussions on difficult concepts and additional resources for areas needing improvement. This ensured learners remained neither bored nor overwhelmed in the learning process. By creating personalized and adaptive learning environments, adaptive AI empowers students to regulate their own learning, increasing motivation, enjoyment, and sustained engagement (Zohra Lhafra & Abdoun, 2023).

5.3.2 Emotionally Supportive AI

Emotionally supportive AI positively influences student motivation and emotional engagement through emotional recognition, empathetic interaction, and gamified encouragement. These impacts are commonly implemented through tools such as conversational AI agents (Aslan et al., 2024; Huang et al., 2008), AI chatbots (Jeon, 2024; Kong & Yang, 2024; Lee et al., 2023; Zhhexenova et al., 2020), and social robots (Dong, Choi, et al., 2024; Hsu et al., 2022). AI systems can detect students' emotional states, providing supportive feedback or motivational prompts (e.g., motivational phrases such as "Try Again" and "Cheer Up") to help manage frustration and sustain engagement (Lee et al., 2021; Weng et al., 2024). Castellano et al. (2021) illustrated how social robots' encouraging gestures, such as clapping to celebrate task completion, reinforce positive learning behaviors (Castellano et al., 2021). Additionally, gamified AI tools, such as interactive math games, integrate game elements to simulate enjoyment and curiosity (Aslan et al., 2024; Burak & Gultekin, 2024). This emotionally supportive approach fosters a positive learning atmosphere, making students feel encouraged, confident, and motivated to persist in learning tasks.

5.3.3 Collaborative and Interactive AI

Collaborative and interactive AI technologies enhance motivation and engagement by simulating authentic social interaction, often through tools like generative AI (Tai & Chen, 2024), AI chatbots (Dong, Mitchell, et al., 2024; Lee et al., 2023; Zhhexenova et al., 2020), and social robots (Ferreira et al., 2020; Hsu et al., 2022; Mubin et al., 2019). These AI systems often function as role-playing partners, such as acting as teaching assistants (Dong, Mitchell, et

AI ON MOTIVATION AND ENGAGEMENT

al., 2024; Ward et al., 2013), language learners (Zhhexenova et al., 2020), storytellers (Ferreira et al., 2020), or game competitors (Castellano et al., 2021), to create realistic collaborative contexts. For example, YiTing et al. (2008) demonstrated AI's ability to engage students in reciprocal interactions, transforming traditional learning activities into dynamic, social experiences, thereby increasing their engagement. The integration of online forums within AI-powered learning platforms further facilitates student-to-student and student-to-teacher interactions, promoting collaboration, peer recognition, and agentic engagement (Lee et al., 2023; Song et al., 2023). Overall, collaborative AI environments increase students' sense of connectedness, fostering deeper engagement and sustained motivation by simulating authentic social experiences and encouraging students to actively participate and collaborate.

5.3.4 Immersive and Multimodal AI

Immersive and multimodal AI, such as generative AI (Chen et al., 2024; Jauhiainen & Garagorry Guerra, 2024; Tai & Chen, 2024), can generate and present content in various formats such as text, images, audio, and animations. AI can be further integrated with Virtual Reality (VR) and Augmented Reality (AR) to create rich, multisensory, immersive learning experiences, sustaining student engagement and interest (Cheng et al., 2024; Qian et al., 2023; Song et al., 2023). For example, Lee et al. (2021) described a science-teaching robot employing animations, digital photographs, and dynamic visual effects to enhance students' understanding of complex concepts, thereby increasing motivation and participation (also see Kong & Yang, 2024). Furthermore, AI-powered AR/VR applications provide highly immersive experiences, allowing students to actively explore knowledge through interactive, hands-on simulations (Aslan et al., 2024). For instance, Song et al. (2023) illustrated this approach by integrating AI with VR for vocabulary instruction, where VR provided immersive contexts for embodied learning, and AI facilitated object and location recognition, enabling students to freely explore vocabulary within a virtual panoramic environment. (Song et al., 2023). Overall, these immersive and multimodal interactions enhance deep focus, sustain student interest, and deepen engagement by offering authentic, context-rich learning experiences (Cheng et al., 2024).

AI ON MOTIVATION AND ENGAGEMENT

5.3.5 *Feedback-oriented AI*

Feedback-oriented AI tools, such as automated evaluation systems (Wilson et al., 2021; Wilson et al., 2024) and generative AI feedback (Tai & Chen, 2024; Villan & dos Santos, 2023; Zheng & Tse, 2023) can support students in monitoring progress, correcting errors promptly, and enhancing their learning engagement and motivation. For example, Ward et al. (2013) presented an intelligent tutoring system that analyzed students' verbal responses to assess their understanding and generated follow-up questions based on their answers, guiding them toward constructing accurate explanations of the study topic. This immediate, targeted feedback helps students promptly address misconceptions and improve performance, thus maintaining active engagement and persistence. For example, Aslan et al. (2024) reported that a social robot provided prompting feedback and helped students adjust problem-solving strategies, which minimized frustration and encouraged active engagement. Additionally, AI-enabled dashboards allow teachers and students to visualize progress clearly, identifying areas requiring support and intervention, further enhancing student engagement and motivation (Lee et al., 2023; Wen et al., 2024). Overall, feedback-oriented AI cultivates a supportive and progressive learning environment to strengthen student engagement and motivation through timely, personalized guidance and clear visualization of learning progress.

5.4 RQ4: Differential Effects of AI

5.4.1 *Subject Area Differences*

Given the uneven distribution in the number of studies across different subjects, this review focuses on comparing STEM disciplines with Human and Social Disciplines, excluding Interdisciplinary Education, which is represented by only one study. As seen in Figure 5, emotional engagement and intrinsic motivation are the two most frequently mentioned factors in both categories, indicating that AI tools consistently enhance learning enjoyment and students' interest regardless of subject area.

Within STEM disciplines, competence is the third most frequently cited factor, with the effectiveness of AI tools in fostering student confidence and self-efficacy when solving

AI ON MOTIVATION AND ENGAGEMENT

mathematical and scientific problems. In contrast, relatedness is more frequently emphasized in Human & Social Disciplines, likely due to AI's role as virtual or simulated dialogue partners in areas such as language learning. These AI interactions effectively cultivate companionship and facilitate social interaction among students, enhancing learners' sense of relatedness.

5.4.2 Grade-level Differences

Eight studies explored the grade-level differences in students' motivation and engagement with AI-supported learning. Results were mixed, with some studies highlighting significant differences across grades and others reporting no significant results. Generally, findings suggest that lower-grade students tend to demonstrate higher levels of behavioral engagement and intrinsic motivation than their higher-grade peers. Specifically, Wilson et al. (2021) found that a higher proportion of fifth-grade students completed shorter-duration (1–12 minutes) AI-delivered interactive skill-building lessons compared to fourth-grade students (19% vs. 13%; $p = .011$). Conversely, fourth-grade students were more likely to engage in longer AI interactions (13 minutes or more) compared to fifth-graders (28% vs. 21%; $p = .026$). Regarding intrinsic motivation, third-grade students ($M = 3.40$, $SD = 1.37$) showed greater willingness for future intention than fourth-grade ($M = 3.31$, $SD = 1.43$) and fifth-grade students ($M = 3.21$, $SD = 1.36$), suggesting a decline in intrinsic motivation as grade levels increased (Wilson et al., 2021). Similarly, Lee et al. (2021) found that fifth-grade students (92%) were more willing to participate in AI-assisted courses than sixth-grade students (69%). Sixth-grade students expressed reluctance due to a preference for more unstructured free time during class and perceptions of overly complex content (Lee et al., 2021). These results collectively indicate that with primary students advancing through grade levels, their engagement with and motivation for AI-supported learning tend to decrease.

5.4.3 Gender Differences

Only one study explicitly explored gender differences related to AI-supported learning. Zhexenova et al. (2020) explored gender differences in students' general engagement and intrinsic motivation during learning activities with social robots. The study found no

AI ON MOTIVATION AND ENGAGEMENT

significant gender differences in either engagement ($F = 1.09$; $p > 0.05$) or intention to use the robots in future activities ($F = 0.02$; $p > 0.05$) (Zhexenova et al., 2020). Regarding emotional engagement, boys reported slightly higher enjoyment than girls, though the difference was not statistically significant (Zhexenova et al., 2020). Similarly, they found comparable emotional responses before and after robot interaction across genders (Zhexenova et al., 2020). There is also an absence of gender differences in programming self-efficacy, with boys and girls demonstrating similar levels of logical thinking, control, or debugging (all $p > 0.05$), indicating comparable levels between boys and girls (Zhexenova et al., 2020).

5.4.4 Generative vs. Traditional AI

Five studies were identified employing generative AI (Chen et al., 2024; Jauhiainen & Garagorry Guerra, 2024; Tai & Chen, 2024; Zheng & Tse, 2023), revealing its distinctive impacts on motivation and engagement. Specifically, generative AI demonstrates unique capabilities in shaping students' perception of learning value. For example, Zheng and Tse (2023) found that feedback from generative AI in mathematics reinforced students' sense of achievement and helped them connect mathematics learning to other subject areas. Moreover, generative AI also elicits students' strategy adjustment. When interacting with these tools, students actively monitor conversations, assess the effectiveness of their questions, and adjust their linguistic strategies as needed (Tai & Chen, 2024). This adaptive communication reflects a high level of agentic engagement, surpassing the passive responses typically observed with other AI tools.

However, generative AI introduces new challenges, particularly in the emotional engagements. Tai and Chen (2024) reported that approximately 25% of individual-mode users and 17.86% of paired-mode users experienced frustration due to the overly linguistic complexity and length of responses. Compared to other AI tools, while generative AI reduces the rigidity of traditional systems, its complexity and open-ended dialogue may impose greater cognitive load and interactional uncertainty.

6. Discussion

AI ON MOTIVATION AND ENGAGEMENT

This systematic review aimed to examine the impact of AI on primary school students' learning motivation and engagement by synthesizing both quantitative and qualitative evidence across 33 empirical studies. Our findings revealed that AI generally enhances students' motivation and engagement across multiple dimensions, though negative outcomes emerged when AI tools were poorly designed or mismatched with students' developmental needs. Additionally, differential impacts emerged across grade levels, subject areas, and types of AI technologies employed. The following sections further explore these findings.

6.1 RQ1: Discussion of AI's Impact on Motivation

Regarding the three basic psychological needs, AI has demonstrated positive effects on primary school students' autonomy, competence, and relatedness, suggesting that it can help fulfill their basic psychological needs and enhance learning motivation (Aslan et al., 2024; Weng et al., 2024). However, research on autonomy remains limited in both quantitative and qualitative studies, which may be because primary school students typically have low self-directed learning abilities and are more reliant on external guidance (Gilmore, 2018). With AI assistance, they are more likely to rely on technology-driven task recommendations or personalized guidance rather than actively setting goals and planning their own learning paths. In contrast, relatedness has received greater attention. AI, functioning as a learning companion, provides para-social experiences that help fulfill students' need for social interaction (Zhhexenova et al., 2020). As a result, studies have found that students widely acknowledge AI's role in social interaction, team cooperation, and comfortable communication experiences (Mubin et al., 2019; Tai & Chen, 2024). Although quantitative studies primarily measured self-efficacy and qualitative research emphasized student-reported confidence, the empirical evidence consistently shows that AI can effectively boost students' self-beliefs in learning.

Quantitative and qualitative findings suggest that AI impacts student motivation across various dimensions, including general motivation, intrinsic motivation, identified regulation, external regulation, and amotivation. Many studies reported positive effects, indicating that AI can enhance learning interest and recognition of learning tools and content, thereby boosting motivation (Chang & Chen, 2022; Chen et al., 2024; Cheng et al., 2024; Lee et al., 2023).

AI ON MOTIVATION AND ENGAGEMENT

However, some students reported decreased motivation due to difficulties in using AI tools or limited perceived appeal, highlighting the role of individual differences in preferences for human interaction or in coping with technical challenges (Jeon, 2024; Moltudal et al., 2020). Notably, no studies addressed introjected or intrinsic regulation. The absence of research on introjected regulation might be attributed to AI's predominantly supportive rather than evaluative or competitive role, resulting in less internalized pressure from social evaluation compared to human-led environments (Weng et al., 2024). As for intrinsic regulation, current AI tools focus on knowledge delivery and immediate feedback, with less emphasis on self-reflection or motivational training. Intrinsic regulation involves advanced self-regulation, self-reflection, and metacognitive skills, which primary school students may not have fully developed or find difficult to articulate. As a result, researchers tend to focus on more observable motivational outcomes, such as interest and enjoyment, rather than deeper regulatory mechanisms.

6.2 RQ2: Discussion of AI's Impact on Engagement

Both quantitative and qualitative research indicate that AI positively impacts primary school students' general engagement, behavioral engagement, cognitive engagement, and emotional engagement. This suggests that AI, as a learning support tool, can enhance student involvement at multiple levels, whether by increasing learning time, improving attention, or providing positive emotional experiences such as enjoyment and satisfaction (Burak & Gultekin, 2024; Dong, Choi, et al., 2024; Wang et al., 2024). However, some studies reported negative outcomes in emotional engagement, including frustration and boredom, highlighting potential drawbacks when AI interactions become repetitive, mechanical, or overly challenging (Jeon, 2024; Tai & Chen, 2024).

Importantly, agentic engagement—characterized by proactive and self-directed behaviors—appeared exclusively in qualitative studies. Qualitative findings indicated that students frequently demonstrated agentic engagement by actively leveraging additional digital resources (e.g., online dictionaries and search engines) to support their interactions with AI tools (Burak & Gultekin, 2024; Jeon, 2024; Tai & Chen, 2024), posing higher-order questions

AI ON MOTIVATION AND ENGAGEMENT

to AI-based partners (Dong, Choi, et al., 2024), and strategically adjusting communication methods in response to real-time AI feedback (Tai & Chen, 2024). Its absence from quantitative research likely results from the dynamic and interactive nature of student-AI interactions, which are challenging to quantify using standardized measures. Qualitative approaches, such as classroom observations and student interviews, effectively captured these nuanced, context-specific interactions, providing insights into how students utilize AI features to actively shape their learning experiences (Reeve et al., 2025b). This methodological discrepancy suggests the need for future studies to adopt mixed-method approaches that can better assess agentic engagement within dynamic AI-supported environments.

6.3 RQ3: Discussion on Types of AI Tools Affecting Motivation and Engagement

Analyzing AI design features, this review categorized effective AI tools into five types: Personalized and Adaptive AI, Emotionally Supportive AI, Collaborative and Interactive AI, Immersive and Multimodal AI, and Feedback-Oriented AI. Each type uniquely contributes to fostering supportive learning environments, enhancing students' engagement and motivation from multiple dimensions (Galindo-Domínguez et al., 2025; Grassini, 2023). Importantly, practical applications suggest that highly effective AI learning tools often integrate several of these features simultaneously (Ferreira et al., 2020; Hsu et al., 2022; Stone, 2023). Numerous empirical studies within this review validated that combining multifunctional design features significantly enhances overall student enthusiasm, sustained engagement, and positive attitudes toward learning (Castellano et al., 2021; Lee et al., 2023; Song et al., 2023). Thus, future AI tool development and implementation should strategically integrate these complementary design features to maximize educational effectiveness and student motivation.

6.4 RQ4: Discussion on the Differential Effects of AI

The differential effects of AI suggest that the impact of AI tools on primary school students' learning motivation and engagement varies according to the subject area and grade level, indicating the necessity of tailoring AI integration to learner-specific characteristics and contexts.

AI ON MOTIVATION AND ENGAGEMENT

The impact of AI varies significantly across subject areas. In STEM disciplines, the most prominent outcome is an enhanced sense of competence, as AI tools support students in building confidence and self-efficacy in addressing complex mathematical and scientific problems (Chettaoui et al., 2020; Weng et al., 2024). In contrast, relatedness received greater emphasis in humanities and social sciences, particularly in language learning (Tai & Chen, 2024). This may be because AI often serves as virtual or simulated dialogue partners, fostering a stronger sense of companionship and promoting student social interaction (Castellano et al., 2021; Zhexenova et al., 2020). These disciplinary differences underscore the importance of aligning AI tool design and implementation with the unique goals and needs of subject-specific contexts to ensure meaningful and effective integration (Schleiss et al., 2024).

Grade-level difference indicates that younger students exhibit higher levels of behavioral engagement and intrinsic motivation compared to their older peers (Lee et al., 2021; Wilson et al., 2021). This pattern can be understood through individual developmental perspectives, which suggest that younger students tend to possess greater curiosity and openness to novel experiences, making them more likely to perceive AI tools as enjoyable and exciting to stimulate active learning behaviors. In contrast, older students may experience increased academic pressure or develop more critical attitudes toward technology use, resulting in reduced willingness to engage with AI-assisted learning (Lee et al., 2021). Therefore, these findings suggest that instructional strategies should be adapted to the developmental stages, providing targeted motivational support tailored specifically to the needs and preferences of each age group. (Xie & Luo, 2025).

Lastly, only one study specifically examined gender differences in students' behavioral engagement, emotional responses, or self-efficacy in interactions with AI (Zhexenova et al., 2020). This single study found no significant gender disparities, suggesting that AI may provide a relatively equitable learning environment at the primary school level (Bao et al., 2024; Mittal et al., 2024). However, due to the limited scope and generalizability of findings based on a single study, future research should leverage larger, diverse, and representative samples to further explore potential gender differences. Additionally, despite the absence of significant gender differences in that limited evidence, it remains important to embed gender

AI ON MOTIVATION AND ENGAGEMENT

inclusivity in the design of AI tools to foster more equitable and supportive learning environments for all students (Xia et al., 2022).

The findings on generative AI highlight its distinct advantages over traditional AI systems. Unlike traditional AI, which typically focused on task completion and accuracy through fixed-feedback mechanism, generative AI fosters deeper reflection on the broader meaning and relevance of learning (Izadi & Forouzanfar, 2024). By promoting connections across disciplines and enhancing students' sense of purpose, generative AI acts as a catalyst for developing more self-determined forms of motivation (Guo et al., 2024; Shah et al., 2025).

Furthermore, whereas traditional AI often limits interaction through preset menus, closed question banks, and fixed feedback (Izadi & Forouzanfar, 2024; Stamper et al., 2024), generative AI's open-ended, dynamic dialogue structure promotes greater self-regulation and active participation (Sardi et al., 2025; Wu et al., 2023). Students interacting with generative AI are required to monitor, evaluate, and adapt their communication strategies, fostering higher agentic engagement than traditional AI typically supports.

However, the flexibility of generative AI brings additional cognitive demands, which may overwhelm some students, leading to frustration and reduced emotional engagement. This risk is particularly salient among younger learners, who may find the complex and unpredictable responses challenging to navigate.

These findings suggest that there is a need for pedagogical support and technical calibration to optimize the benefits of generative AI. Careful instructional design is needed to ensure that generative AI enhances, rather than detracts from, students' motivation and engagement, particularly in primary education settings.

7. Limitations, Recommendations, and Conclusion for Future Studies

7.1 Limitations of the included studies

First, while many studies referenced motivation and engagement frameworks, their primary research objectives centered on evaluating AI tool functionality and effectiveness rather than systematically investigating psychological constructs. This limitation has led to conceptual and measurement inconsistencies, such as unclear distinctions between related

AI ON MOTIVATION AND ENGAGEMENT

concepts like intrinsic motivation and emotional engagement. These inconsistencies have, to some extent, undermined the robustness and comparability of the findings. Future empirical research should place greater emphasis on students' motivation and engagement as core research variables, grounded in established psychological and educational theories, and adopt a multidimensional perspective to facilitate accurate comparisons and synthesis across studies.

Second, although 27 of the 33 included studies used quantitative methods, a meta-analysis was not feasible due to marked heterogeneity in research designs and outcome measures. Some studies were observational, others lacked control groups, pretests, or randomization, and only six were RCTs. These inconsistencies limited the ability to calculate standardized, comparable effect metrics across studies.. Future studies should adopt more rigorous and standardized designs—particularly RCTs—to enable robust meta-analytic synthesis.

Third, the included studies showed several methodological limitations that may affect the credibility of the findings. One issue is that only one study conducted prospective registration, while it was absent in the rest, increasing the risk of selective reporting and reducing result transparency. Another issue is that most studies did not implement blinding for teachers or researchers, which may have led to differences in instruction or subjective bias in evaluation, thus affecting the fairness and reliability of the results. In addition, seven studies used quasi-experimental designs without random assignment, which may have introduced systematic differences between the intervention and control group, undermining the internal validity of the findings. These issues highlight the need for future studies to adopt more rigorous practices, including prospective registration, randomized experiments, and the use of blinding and standardized reporting to enhance research quality and credibility.

Fourth, most studies employed cross-sectional or short-term research designs, limiting insights into the developmental trajectories and long-term impacts of AI on student motivation and engagement. To overcome this limitation, longitudinal or experimental research designs are recommended to track the developmental trajectories of student motivation and engagement over extended periods, providing clearer insights into developmental patterns and sustained effects of AI-supported learning.

Last, individual differences in student responses to AI tools, such as gender, grade level,

AI ON MOTIVATION AND ENGAGEMENT

prior technology experience, and socioeconomic background, were rarely comprehensively addressed. Future studies should systematically examine how these individual characteristics influence motivation and engagement with AI, leveraging diverse and representative samples to identify tailored educational strategies that effectively support varied learner populations.

7.1 Limitations of this review

While this review provides valuable insights into the impact of AI on primary school students' learning motivation and engagement, several limitations remain, offering directions for future research.

First, the literature search in this review was limited to major academic databases and English-language publications. This may have excluded relevant studies in other languages or from less widely indexed sources, resulting in a partial view of the current research landscape. Future studies should expand the scope of literature sources and languages, incorporating findings from multilingual publications beyond mainstream English-language databases. A more inclusive literature search strategy could provide a more comprehensive view of the research progress on how AI influences primary school students' motivation and engagement.

Second, this review applied SDT to synthesize and interpret findings on primary students' motivation. Future research could incorporate diverse theoretical perspectives or compare multiple frameworks to deepen understanding and identify complementary insights. Besides, specific motivational dimensions such as introjected and integrated regulation remain underexplored, particularly among younger primary school students. Given that these motivational aspects involve nuanced internal processes that may be challenging for younger students to articulate, future research should employ age-appropriate methodologies, such as observational techniques or age-sensitive self-report instruments, to investigate these motivational dimensions effectively.

7.2 Conclusion

This systematic review synthesized 33 empirical studies and, guided by the SDT and multidimensional engagement frameworks, examined the impact of AI on primary school

AI ON MOTIVATION AND ENGAGEMENT

students' learning motivation and engagement. The findings indicate AI positively influences students' general motivation, intrinsic motivation, and two subtypes of extrinsic motivation — external and identified regulation, while effectively fulfilling their basic psychological needs for autonomy, competence, and relatedness. AI also demonstrated positive effects across behavioral, cognitive, emotional, and agentic dimensions, underscoring its capacity to foster active participation and sustained learning interest. However, some studies identified inconsistent or negative outcomes, such as increased amotivation, reduced curiosity, and lower intentions for future engagement, particularly when AI tools were poorly designed or misaligned with developmental needs. The review also found differential effects across subject areas, grade levels, and, to a lesser extent, gender.

In conclusion, thoughtfully designed AI tools integrating multiple key design features such as personalized adaptivity, emotional support, collaborative interaction, immersive multimodal experiences, and feedback mechanisms can effectively support learning environments, offering promising potential in sustaining and enhancing primary school students' motivation and engagement.

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AI ON MOTIVATION AND ENGAGEMENT

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AI ON MOTIVATION AND ENGAGEMENT

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AI ON MOTIVATION AND ENGAGEMENT

Figure 1

The Self-determination Theory Framework

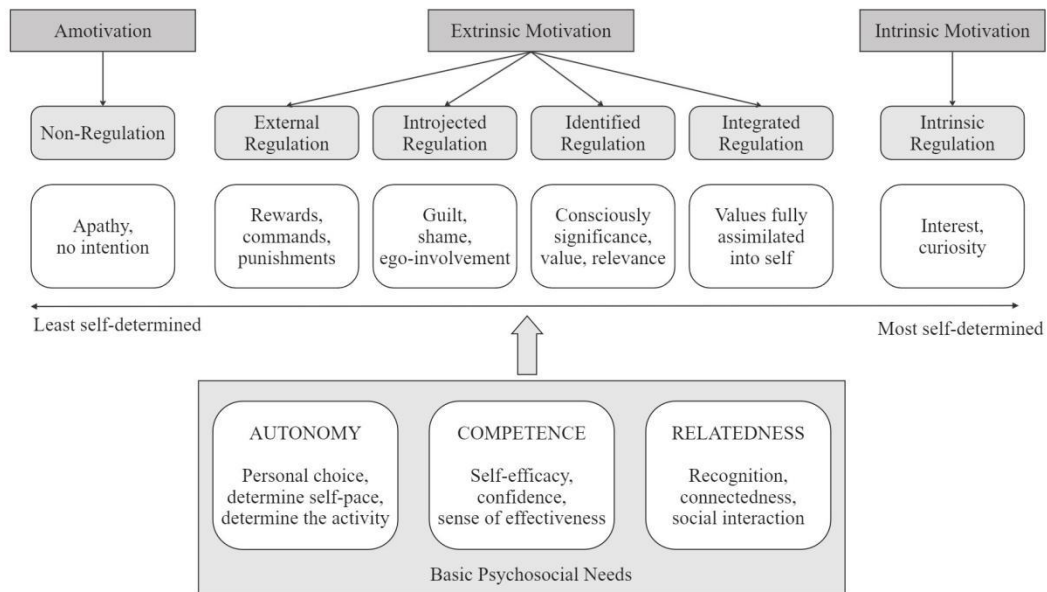
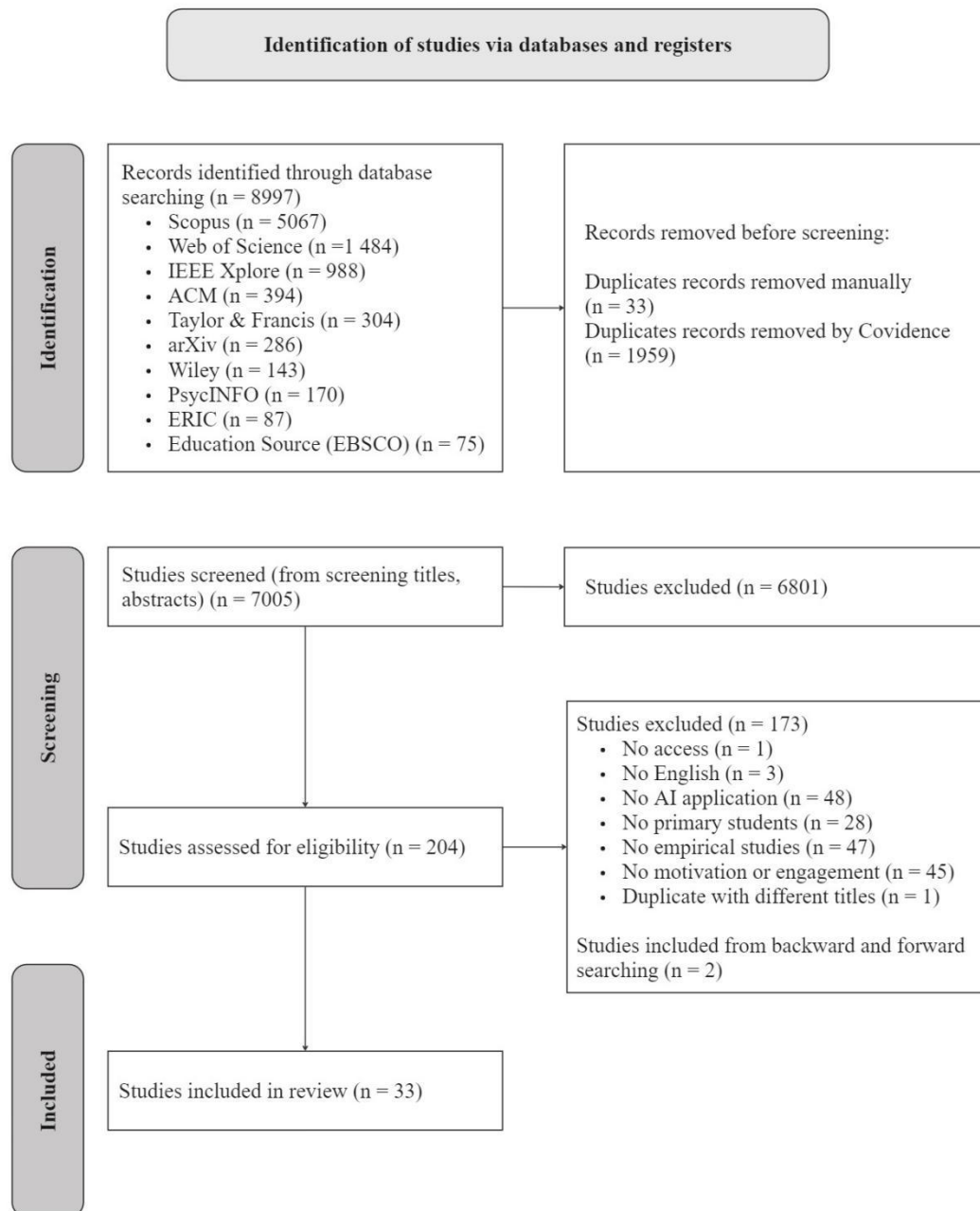


Figure 2

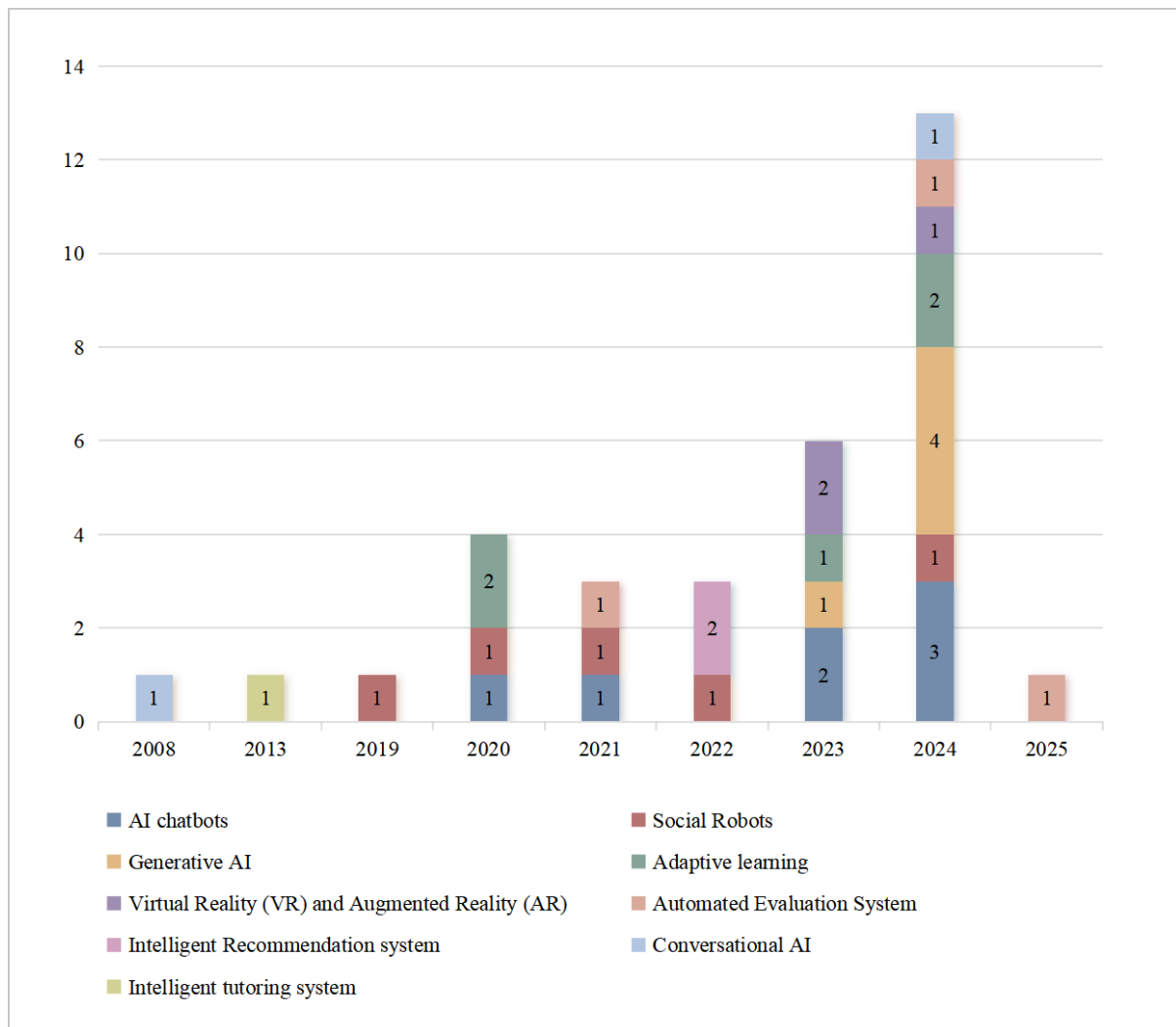
PRISMA Flow Diagram of Article Selection



AI ON MOTIVATION AND ENGAGEMENT

Figure 3

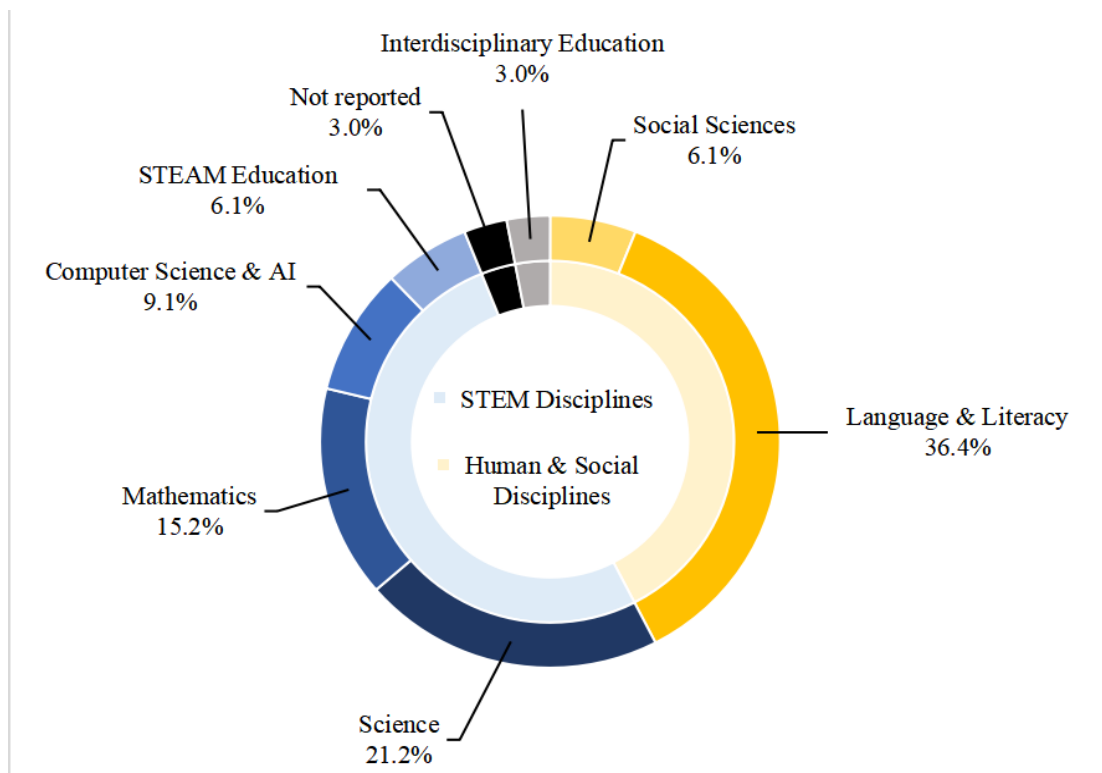
Number of AI Tools Types



AI ON MOTIVATION AND ENGAGEMENT

Figure 4

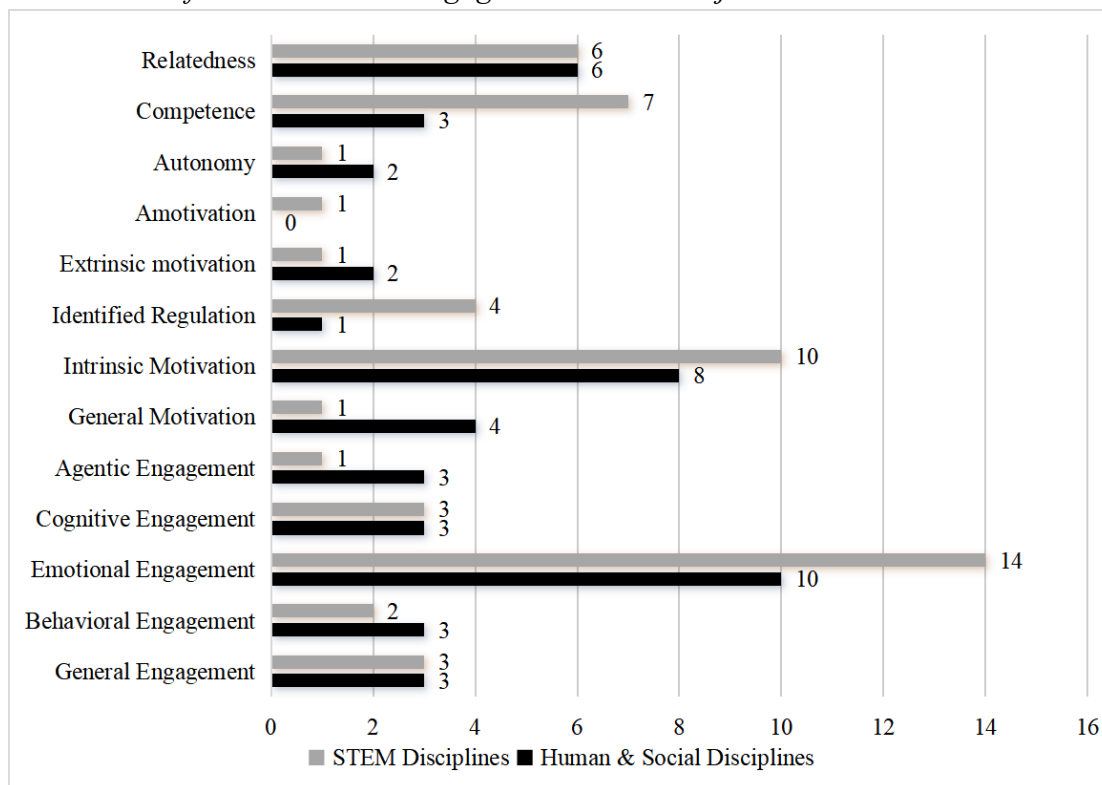
Number of Studies by Subject



AI ON MOTIVATION AND ENGAGEMENT

Figure 5

Distribution of Motivation and Engagement Across Subject Areas



AI ON MOTIVATION AND ENGAGEMENT

Table 1

Inclusion and Exclusion Criteria

	Inclusion criteria	Exclusion criteria
Population	- Primary school students (typically aged 4-12), including different genders, grades, and learning abilities. - Normal (non-disabled) students.	- Non-primary school populations (e.g., preschool children, middle school students or adults) or studies that do not clearly specify the age range of participants. - Students with disabilities.
Exposure	The application of AI	Not involving AI or discussing technologies that are not AI (e.g., traditional teaching methods or non-AI technologies or tools).
Context	- Conducted in educational settings - Contexts of AI's practical application on primary school students' motivation.	- Unrelated to educational settings (e.g., commercial, medical fields) - Irrelevance to AI's impact on primary school students' motivation (e.g., students' health or exercises).
Outcome	Learning motivation and related concepts such as self-efficacy, achievement goals, self-determination, or engagement.	Lack of focus on learning motivation or related concepts (e.g., motivation for using technology/AI).
Types of studies	Any literature presenting original data.	Literature not presenting original data (e.g., reviews and meta-analysis).
Language	Written in English	All other languages
Time	No time constraints	-

Table 2*Extracted Information of the Included Studies*

Authors	Country/Region	Technology	Learning Subject	Sample Characteristics	Grade	Research Method
Aslan et al. (2024)	the United States	Conversational AI	Mathematics	14 students (8 girls, 6 boys)	1st grade	Mixed research method
Burak and Gultekin (2024)	Turkey*	Adaptive learning	Social Sciences	36 students (24 boys, 12 girls) (Mage = 9.8 years, SD = 0.32)	4th grade	Qualitative method
Castellano et al. (2021)	Italy	Social robots	Science	51 students (26 boys, 25 girls) (age = 7–9 years)	N/A	Quantitative method
Chang and Chen (2022)	China	Intelligent Recommendation system	Computer Science & AI	Approximately 252 students	N/A	Quantitative method
Chen et al. (2024)	China	Generative AI	Language & Literacy	80 students (experimental: 40 [20 boys, 20 girls]; control: 40 [21 boys, 19 girls])	5th grade	Mixed research method
Cheng et al. (2024)	the United States*	Augmented Reality (AR)	Science	50 students (31 boys, 19 girls) (Grades 2–5: 12% second, 32% third, 32% fourth,	2nd to 5th grade	Quantitative method

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AI ON MOTIVATION AND ENGAGEMENT

Chettaoui et al. (2020)	Tunisia*	Adaptive learning	Science	24% fifth) 18 students (age = 9–10 years)	4th grade	Quantitative method
Dong, Choi, et al. (2024)	the United States	Social Robots	STEAM education	11 students (6 boys, 5 girls) (Mage = 9.76 years, SD = 0.81)	3rd to 5th grade	Mixed research method
Dong, Mitchell, et al. (2024)	the United States	AI chatbots	STEAM education	30 students (16 girls, 14 boys) (age = 5.42–11.92 years, M = 8.04, SD = 1.79).	N/A	Mixed research method
Feng and Wang (2023)	China	AI chatbots	Language & Literacy	85 students (AI-assisted: 43 [22 boys, 21 girls]; paper book: 42 [22 boys, 20 girls]) (Mage_AI-assisted = 11.26 years, SD = 0.36; Mage_paper book = 11.29 years, SD = 0.31)	5th grade	Mixed research method
Ferreira et al. (2020)	Madeira, Portugal*	Social Robots	Science	105 students (56 boys, 49 girls) (Grade 1: 29, Grade 2: 25, Grade 3: 26, Grade 4: 25) (age = 6–10 years).	1st to 4th grade	Mixed research method
Hsu et al. (2022)	Singapore	Social Robots	Interdisciplinary Education	46 students (26 boys, 20 girls)	5th grade	Quantitative method

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AI ON MOTIVATION AND ENGAGEMENT

Huang et al. (2008)	China*	Conversational AI	Language & Literacy	34 students (18 boys, 16 girls)	5th grade	Mixed research method
Jauhiainen and Garagorry Guerra (2024)	Uruguay	ChatGPT	Social Sciences	110 students (4th grade: 50 [57.1% boys, 42.9% girls]; 5th grade: 32 [68.7% boys, 31.3% girls]; 6th grade: 28 [46.4% boys, 46.4% girls, 7.1% undisclosed]) (age = 8–14 years).	4th to 6th grade	Quantitative method
Jeon (2024)	South Korea	AI chatbots	Language & Literacy	36 students	N/A	Qualitative method
Kong and Yang (2024)	Hong Kong	AI chatbots	Computer Science & AI	87 students (56.3% boys, 43.7% girls) (age = 9–10 years)	5th grade	Mixed research method
Lee (2024)	South Korea	Adaptive learning	Mathematics	23 students (13 girls, 10 boys)	3rd grade	Mixed research method
Lee et al. (2021)	China	AI chatbots	Computer Science & AI	281 students	5th to 6th grade	Mixed research method
Lee et al. (2023)	South Korea	AI chatbots	Science	192 =students (experimental: 81 [33 boys, 48 girls]; control: 111 [58 boys, 53 girls]).	6th grade	Quantitative method

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AI ON MOTIVATION AND ENGAGEMENT

Moltudal et al. (2020)	Norway	Adaptive learning	Mathematics	43 students (age = 10–12 years)	5th to 7th grade	Quantitative method
Mubin et al. (2019)	United Arab Emirates (UAE)	Social Robots	Mathematics	44 students (robot condition: 24; human confederate: 20) (age = 10–12 years).	4th grade	Mixed research method
Qian et al. (2023)	China	Augmented Reality (AR)	Language & Literacy	14 students (7 boys, 7 girls) (age = 8–9 years)	3rd grade	Mixed research method
Song et al. (2023)	Hong Kong	Virtual Reality (VR) and Augmented Reality (AR)	Language & Literacy	15 students (8 girls, 7 boys) (age = 6–7 years)	1st grade	Mixed research method
Tai and Chen (2024)	China	Generative AI	Language & Literacy	85 students (I-Bot: 28 [16 boys, 12 girls]; P-Bot: 28 [15 boys, 13 girls]; No-Bot: 29 [14 boys, 15 girls]) (Mage_I-Bot = 12.32; Mage_P-Bot = 13.13; Mage_No-Bot = 12.01).	6th grade	Mixed research method
Wang et al. (2022)	China	Intelligent Recommendation system	Language & Literacy	35,522 students (pilot: 16,896; non-pilot: 16,626) (responses: pilot = 12,902, non-pilot = 11,754)	N/A	Quantitative method
Ward et al. (2013)	the United States	Intelligent tutoring system	Science	1,478 students across 22 schools and 63 classrooms (MyST tutoring: 102; human tutoring: 85; control: 1,155)	N/A	Quantitative method

AI ON MOTIVATION AND ENGAGEMENT

Wen et al. (2025)	Singapore	Automated Evaluation System	Language & Literacy	140 students	2nd grade	Mixed research method
Weng et al. (2024)	China	Generative AI	Science	133 students (experimental: 67 [33 boys, 34 girls]; control: 66 [34 boys, 32 girls])	5th grade	Quantitative method
Wilson et al. (2021)	the United States	Automated Evaluation System	Language & Literacy	1935 students (955 boys, 980 girls) (Grades 3–5: 668 [34.5%], 747 [38.6%], 520 [26.9%]) from 14 elementary schools	3rd to 5th grade	Quantitative method
Wilson et al. (2024)	the United States	Automated Evaluation System	Language & Literacy	3459 elementary school students (ELs: 820 [457 boys, 363 girls]; non-ELs: 2639 [1294 boys, 1345 girls]) (Grades 3–5: 3rd = 1123 [32.5%], 4th = 1187 [34.3%], 5th = 1149 [33.2%])	3rd to 5th grade	Quantitative method
Zheng and Tse (2023)	Hong Kong	Generative AI	Mathematics	21 students	4th grade	Mixed research method
Zhexenova et al. (2020)	Kazakhstan	AI chatbots	Language & Literacy	62 children (31 boys, 31 girls) (age = 7–9 years).	2nd to 3rd grade	Quantitative method
Zohra Lhafa and Abdoun (2023)	Morocco*	Adaptive learning	Interdisciplinary Education	30 students	N/A	Quantitative method

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AI ON MOTIVATION AND ENGAGEMENT

Table 3*Quantitative Findings on Psychological Needs*

Authors	Research Design	Research Method	Category	Findings	Effect and Significance
Autonomy					
Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Autonomy	Autonomy: Autonomy remained unchanged from pre (M = 3.69, SD = 0.62) to post (M = 3.75, SD = 0.61) (Mdiff = 0.06, SD = 0.51, ES = 0.05, p = 0.450).	No significant
Competence					
Lee et al. (2023)	Quasi-experimental (pre/post & control)	MANOVA	Competence	Confidence: The experimental group showed greater increase for confidence in science (Mean $\pm$ SD = experimental, 1.81 ± 14.46) than the control (Mean $\pm$ SD = 1.60 ± 9.81), with no significant difference. Positive responses for being good at science increased more in the experimental group than in the control (5.2% vs. 3.2%).	No significant
Weng et al. (2024)	RCT	ANCOVA	Competence	Self-Efficacy: The experimental group reported higher self-efficacy (M = 3.57/5, SD = 0.884) (Adjusted scores M = 3.57/5, SD = 0.884) than the control group (M = 3.27/5, SD = 0.932) (Adjusted scores M = 3.27/5, SD = 0.932) (F = 6.857, p = 0.010, n2=0.234).	Positive Significant
Qian et al. (2023)	RCT	ANCOVA	Competence	Self-efficacy: Children learning with IARE showed a higher average change (0.29) than those learning with flashcards (-0.51) (F = 33.354, p < 0.01).	Positive Significant

AI ON MOTIVATION AND ENGAGEMENT

Hsu et al. (2022)	Quasi-experimental (pre/post & control)	Paired sample t tests	Competence	Self-Efficacy: Both boys (pre-test: 8.92 to post-test: 12.04) (boys: $t = -3.92$, $p < 0.01$, $ES = 1.03$) and girls (pre-test: 8.13 to post-test: 11.66) (girls: $t = -5.42$, $p < 0.001$, $ES = 1.30$) showed improvements in self-efficacy programming.	Positive Significant
Castellano et al. (2021)	Pre-experimental	ANOVA	Competence	Self-Efficacy: Students reported a significant increase in perceived efficacy from pre ($M = 2.47$) to post ($M = 2.71$) ($F(2,48) = 5.79$, $p < 0.05$).	Positive Significant
Chettaoui et al. (2020)	RCT	Descriptive Statistics	Competence	Self-efficacy: Students showed a greater sense of efficacy when manipulating physical tangible objects ($M = 4.78/5$, $SD = 0.44$) compared to learning with tablets ($M = 3.77/5$, $SD = 1.09$).	Positive
Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Competence	Competence: Competence remained unchanged from pre ($M = 3.73$, $SD = 0.52$) to post ($M = 3.85$, $SD = 0.61$) ($Mdiff = 0.12$, $SD = 0.63$, $ES = 0.28$, $p = 0.186$).	No significant
Relatedness					
Mubin et al. (2019)	Quasi-experimental (post & control)	ANOVA	Relatedness	Social Interaction: Students in the robot condition talked significantly more with their classmates ($F(1,42) = 6.16$, $p = 0.02$), averaging 0.4% of the session duration compared to 0.05% in the human condition.	Positive Significant
Aslan et al. (2024)	Pre-experimental	Paired samples T-test	Relatedness	Social Interaction: Students predominantly demonstrated non-verbal interactions with Oscar (45.7%) and the instructional assistant (43.4%).	Positive
Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Relatedness	Social Interaction: Students reported CoolE Bot providing reciprocal encouragement and support (I-Bot: 75%, P-Bot: 64.29%).	Positive

AI ON MOTIVATION AND ENGAGEMENT

Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Relatedness	Team Cooperation: Students from the P-Bot group reported a preference for interacting with CoolE Bot with a partner (N = 20, 71.43%).	Positive
Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Relatedness	Communication Experience: Students reported CoolE Bot creating a non-judgmental and less anxious atmosphere for EFL speaking (I-Bot: 71.43%, P-Bot: 78.57%).	Positive
Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Relatedness	Relatedness: Relatedness remained unchanged from pre (M = 4.46, SD = 0.59) to post (M = 4.43, SD = 0.57) (Mdiff = -0.03, SD = 0.52, ES = 0.03, p = 0.766).	No significant

AI ON MOTIVATION AND ENGAGEMENT

Table 4

Excerpt of Qualitative Findings on Psychological Needs

Category	Findings	
Autonomy	Self-paced Learning	- “Thirteen students stated, “We can now read picture books independently and at our own pace thanks to the AI robot.” (Feng & Wang, 2023)
	Independent Learning	- “Among the 26 students, eight students, including S2 and S11, elaborated more on their preference for the chatbots by citing the independent learning opportunities that the chatbots created.” (Jeon, 2024)
Competence	Confidence	- “It pointed out my talent in mathematics and specifically recorded my achievements, which made me more confident. (Zheng & Tse, 2023)”
Relatedness	Team Cooperation	- “Students in the AI class said that this type of English reading helped them develop team spirit by allowing them to engage in more “group cooperative role-play”. (Feng & Wang, 2023)
	Communication Experience	- “Twenty-six students stated that they felt comfortable talking to the chatbots because they did not feel pressure from having to worry about human partners, specifically, about their speech mistakes; that is, there was no sense of being judged by their peers or teacher.” (Jeon, 2024)
	Social Interaction	- “When AI-generated formative feedback was used to compare with peers and be seen as an expectation of the teacher, minority students felt this aspect gave them more social-emotional interaction.” (Zheng & Tse, 2023)

Table 5*Quantitative Findings on Motivation*

Authors	Research Design	Research Method	Category	Findings	Effect and Significance
General Motivation					
Cheng et al. (2024)	RCT	Paired samples T-test	General Motivation	General Motivation: Students' motivation to learn science significantly increased from pre (M = 21.7, SD = 2.96) to post (M = 22.9, SD = 3.8), (t(47) = -2.43, p = 0.019, d = 0.35).	Positive Significant
Chen et al. (2024)	Quasi-experimental (pre/post & control)	ANCOVA	General Motivation	General Motivation: The experimental group reported higher overall motivation (Adjusted M = 22.67) than the control group (Adjusted M = 21.46) (F = 2.18, p > .05, $\eta^2 = 0.03$).	No significant
Wilson et al. (2024)	Pre-experimental	Linear regression	General Motivation	General Motivation: Students showed lower agreement on MI Write boosting motivation (Non-ELs (M = 3.78/5), EL–Proficient (M = 4.07/5), EL–Non-proficient(M = 3.73/5)).	Negative
Wilson et al. (2021)	Quasi-experimental (pre/post & control)	Descriptive Statistics	General Motivation	General Motivation: Students showed lower agreement on MI Write boosting motivation (M = 3.32).	Negative
Intrinsic Motivation					
Qian et al. (2023)	RCT	ANCOVA	Intrinsic Motivation	Interest: Children learning with IARE showed a higher average change (0.71) than those learning with flashcards (0.09) (F = 17.708, p = 0.001).	Positive Significant

AI ON MOTIVATION AND ENGAGEMENT

Lee et al. (2023)	Quasi-experimental (pre/post & control)	MANOVA	Intrinsic Motivation	Interest: The experimental group showed greater increase (7.80 ± 18.65) than the control group (0.75 ± 15.02) ($F = 8.52, p < 0.01$) after learning with the AI chatbot.	Positive Significant
Cheng et al. (2024)	RCT	Paired samples T-test	Intrinsic Motivation	Interest: More students in the treatment group (32.0%) reported willingness to write more letters than those in the control (8.6%).	Positive Significant
Feng and Wang (2023)	RCT	Independent samples t-test Paired samples T-test	Intrinsic Motivation	Interest: Interest and Motivation: Students reported a 2-level improvement in interest and motivation in English extracurricular reading ("poor" to "relatively good").	Positive Significant
Wang et al. (2022)	Quasi-experimental (post & control)	Chi-square test	Intrinsic Motivation	Interest: More students from pilot schools preferred reading extracurricular books (33.68%) compared to those from non-pilot schools (23.98%) ($p < 0.0001$).	Positive
Chettaoui et al. (2020)	RCT	Descriptive Statistics	Intrinsic Motivation	Interest: Students showed higher interest in learning with tangible interaction ($M = 4.67/5, SD = 0.49$) than with tablets ($M = 3.83/5, SD = 0.79$).	Positive
Dong, Choi, et al. (2024)	Pre-experimental	Descriptive Statistics	Intrinsic Motivation	Interest: Regarding interest in STEAM, 100% (5/5) of the children selected "yes." Regarding interest in the arts, 80% (4/5) selected "yes."	Positive
Jauhiainen and Garagorry Guerra (2024)	Observational	Independent samples t-test	Intrinsic Motivation	Interest: Pupils who strongly liked the test lesson showed consistently high interest in the lesson topic ($t(DF = 16) = 34.198, p = 0.005$).	Positive

AI ON MOTIVATION AND ENGAGEMENT

Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Intrinsic Motivation	Interest: Students expressed a strong interest and willingness to extend their interactions (I-Bot: 67.86%, P-Bot: 57.14%).	Positive
Song et al. (2023)	Observational	Descriptive Statistics	Intrinsic Motivation	Interest: Students reported high engagement in curiosity when using Virtual Go Mode (M=3.47, SD=.39).	Positive
Chen et al. (2024)	Quasi-experimental (pre/post & control)	ANCOVA	Intrinsic Motivation	Interest: The experimental group reported higher intrinsic motivation (Adjusted M = 10.88) than the control group (Adjusted M = 10.70) (F = 0.11, p > .05, $\eta^2 = 0.00$).	No significant
Song et al. (2023)	Observational	Descriptive Statistics	Intrinsic Motivation	Curiosity: Students reported high engagement in curiosity when using Virtual Go Mode (M = 3.67, SD = 0.97).	Positive
Dong, Choi, et al. (2024)	Pre-experimental	Descriptive Statistics	Intrinsic Motivation	Curiosity: Children's curiosity about robots slightly decreased after the program (Mpre = 3.69, Mpost = 3.59).	Negative
Chang and Chen (2022)	Quasi-experimental (post-test & control)	Independent sample t test	Intrinsic Motivation	Future Intention: The teaching video group showed a stronger willingness to continue using the method (M = 3.594 out of 5, SD = 1.283) than the text-based group (M = 3.192 out of 5, SD = 1.155) (p < 0.001).	Positive Significant
Lee (2024)	Observational	Descriptive Statistics	Intrinsic Motivation	Future Intention: 69.4% of students expressed their intention to continue using the AI systems.	Positive
Lee et al. (2021)	Observational	Descriptive Statistics	Intrinsic Motivation	Future Intention: Students (92% of fifth-grade and 69% of sixth-grade) expressed willingness to participate in AI-assisted courses next semester.	Positive
Wilson et al. (2024)	Pre-experimental	Linear regression	Intrinsic Motivation	Future Intention: Students showed lower agreement on future use intent (Non-ELs (M = 3.32/5), EL-Proficient (M = 3.87/5), EL-Non-proficient(M = 3.59/5)).	Negative

AI ON MOTIVATION AND ENGAGEMENT

Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Intrinsic Motivation	Intrinsic Motivation: Students' intrinsic motivation remained unchanged from pre (M = 3.62, SD = 0.88) to post (M = 3.48, SD = 0.89) (Mdiff = -0.14, SD = 0.63, ES = 0.15, p = 0.173).	No significant
Identified Regulation					
Chang and Chen (2022)	Quasi-experimental (post-test & control)	Independent sample t test	Identified Regulation	Usefulness Perception: The teaching video group perceived the learning method as significantly more useful (M = 3.751 out of 5, SD = 1.231) than the text-based group (M = 3.267 out of 5, SD = 1.112) (p < 0.001).	Positive Significant
Lee (2024)	Observational	Descriptive Statistics	Identified Regulation	Usefulness Perception: Students reported the AI interface helpful for their learning (M = 4.35 out of 5, SD=0.78). Value Perception: The experimental group showed a larger increase for perceived value of science (Mean ± SD = 4.00 ± 18.19) than the control (Mean ± SD = 2.18 ± 27.22), with no significant difference. Positive responses to importance of being good at science (9.6% vs 5.1%) and learning science to get ahead (11.9% vs 2.1%) increased more in the experimental group than in the control group.	Positive
Lee et al. (2023)	Quasi-experimental (pre/post & control)	MANOVA	Identified Regulation	Value Perception: The experimental group showed a larger increase for perceived value of science (Mean ± SD = 4.00 ± 18.19) than the control (Mean ± SD = 2.18 ± 27.22), with no significant difference. Positive responses to importance of being good at science (9.6% vs 5.1%) and learning science to get ahead (11.9% vs 2.1%) increased more in the experimental group than in the control group.	No significant
Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Identified Regulation	Identified Regulation: Identified regulation remained unchanged from pre (M = 4.25, SD = 0.58) to post (M = 4.15, SD = 0.61) (Mdiff = -0.10, SD = 0.63, ES = 0.16, p = 0.298).	No significant
External Regulation					
Chen et al. (2024)	Quasi-experimental (pre/post & control)	ANCOVA	Extrinsic Regulation	External Regulation: The experimental group reported higher extrinsic motivation (Adjusted M = 11.79 vs. 10.76) (F = 4.72, p < .05, n ² = 0.05)	Positive Significant

AI ON MOTIVATION AND ENGAGEMENT

Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Extrinsic Regulation	External Regulation: External regulation remained unchanged from pre (M = 2.84, SD = 0.61) to post (M = 2.89, SD = 0.59) (Mdiff = 0.04, SD = 0.46, ES = 0.07, p = 0.556).	No significant
Amotivation					
Moltudal et al. (2020)	Pre-experimental	Paired samples T-test	Amotivation	Amotivation: Amotivation significantly increased from pre (M = 1.68, SD = 0.59) to post (M = 1.93, SD = 0.65) (Mdiff = 0.24, SD = 0.72, ES = 0.4, p = 0.039).	Positive Significant

Table 6

Excerpt of Qualitative Findings on Motivation

Category	Findings	
Intrinsic Motivation	Interest	- “All children who completed the post-program survey reported that their interest in STEAM increased after participating in this program.” (Dong, Choi, et al., 2024)
	Curiosity	- “Learning vocabulary is like going on an adventure. I am curious as to what fruits I may discover in various color zones.” (Song et al., 2023)
	Future Intention	- “All 14 students indicated that they wanted to come back as they were excited to do the following with Oscar next time: Plant and water flowers, pick a color they like to plant, play the warm-up game, and play hide and seek next time with Oscar.” (Aslan et al., 2024)
Identified Regulation	Usefulness Perception	- “Many of the students recognized the chatbots as helpful tools from which they could learn more; this perception ultimately led to increased motivation to interact with the chatbots using English.” (Jeon, 2024)
	Value Perception	- “A few students noted that the AI generated feedback on mathematics changed their perceptions of the importance of mathematics, such as affirming individual values and highlighting the links between mathematical performance and other subjects.” (Zheng & Tse, 2023)
Extenal Regulation	External Regulation	- “The researcher observed that getting applause at UYEM and playing games as a reward has a very important reinforcement effect, and getting this reinforcement motivates students and increases their motivation.” “I solved it correctly in the second test, I got applause, and I started the games.” (Burak & Gultekin, 2024)
Amotivation	Amotivation	- “Her descriptions bear some indications of the program reinforcing her feeling of amotivation and incompetence: - It says, "figure this out," but often you don't know what to do.” (Moltudal et al., 2020)
General Motivation	General Motivation	- “The students considered the chatbot tasks in the configuration not as attractive as tasks with human partners. This negative perception toward the chatbots lowered their motivation to conduct language tasks with the chatbots.” (Jeon, 2024)

Table 7*Quantitative Findings on Engagement*

Authors	Research Design	Research Method	Category	Findings	Effect and Significance
General Engagement					
Wilson et al. (2021)	Quasi-experimental (pre/post & control)	Descriptive Statistics	General Engagement	General Engagement: Students' attitudes toward AWE were positive, with mean scores ranging from 3.32 to 4.38 (SD = 1.11), a median of 4 to 5, and a mode of 5 for all items.	Positive
Dong, Choi, et al. (2024)	Pre-experimental	Descriptive Statistics	General Engagement	General Engagement: Children's self-rated engagement ratings were high (M = 3.49 out of 4, SD = 0.29). General Engagement: Children in the Robot condition rated their experience as slightly higher (M = 80.75, SD = 17.71, Mdn = 77.5, Max = 100, Min = 50) than those children in the Tablet (M = 77.5, SD = 18.17, Mdn = 75, Max = 100, Min = 40) and Teacher conditions (M = 76.32, SD = 17.55, Mdn = 75, Max = 100, Min = 45), though not significantly..	Positive
Zhexenova et al. (2020)	RCT	Mann-Whitney U tests	General Engagement	General Engagement: Students reported increased engagement, with a mean increase from 3.71 (SD = 0.69) to 4.01 (SD = 0.73) (Mdiff = 0.31, $p < .001$).	No significant
Kong and Yang (2024)	Pre-experimental	Paired sample t tests	General Engagement	General Engagement: The students receiving adaptive learning activities based on their needs showed consistently higher engagement than the students with randomly assigned activities (nearly 100% vs. 80% and 60% vs. 50% in two problem situations).	Positive Significant
Zohra Lhafa and Abdoun (2023)	Pre-experimental	Descriptive Statistics	General Engagement		Positive

AI ON MOTIVATION AND ENGAGEMENT

Behavioral Engagement					
Wang et al. (2022)	Quasi-experimental (post & control)	Chi-square test	Behavior Engagement	Learning Time: The number of pilot schools with more than 30 minutes reading time daily was 45.94%, nearly 10% higher than non-pilot schools (34.45%) ($p < 0.0001$).	Positive Significant
Feng and Wang (2023)	RCT	Independent samples t-test Paired samples T-test	Behavioral Engagement	Learning Time: The AI class's daily reading time reached 1.86h (9 times pre-experiment levels) and was 1.6 times longer than the paper book class. Total reading time increased dramatically over six months, reaching 334.8h compared to 207h in the paper book class.	Positive
Feng and Wang (2023)	RCT	Independent samples t-test Paired samples T-test	Behavioral Engagement	Learning Volume: The AI class's daily reading reached 10,000 Chinese characters, totaling 1,800,000 characters over the experiment period (20 times pre-experiment levels).	Positive
Wilson et al. (2024)	Pre-experimental	Linear regression	Behavioral Engagement	Learning Volume: Students completed more drafts compared from spring to fall. Non-ELs increased from 1.61 (SD = 0.514) to 1.80 (SD = 0.563), proficient ELs from 1.62 (SD = 0.487) to 1.76 (SD = 0.468), and non-proficient ELs from 1.55 (SD = 0.531) to 1.69 (SD = 0.558).	Positive
Feng and Wang (2023)	RCT	Independent samples t-test Paired samples T-test	Behavioral Engagement	Learning Content: The AI class maintained a 1:1 balance between Chinese and English, while the paper book class shifted to 1:0.2 in favor of Chinese.	Positive

AI ON MOTIVATION AND ENGAGEMENT

Cognitive Engagement					
Chen et al. (2024)	Quasi-experimental (pre/post & control)	ANCOVA	Cognitive Engagement	Higher-order Thinking: After controlling for the pretest, the experimental group scored significantly higher in critical thinking (Adjusted M = 23.38) than the control group (Adjusted M = 20.62) ($F = 7.31, p < .01, \eta^2 = 0.08$).	Positive Significant
Cheng et al. (2024)	RCT	Paired samples T-test	Cognitive Engagement	Higher-order Thinking: The treatment group reported higher AI-encouraged deeper thinking (20% strongly agreed, 36% agreed) than the control (13% strongly agreed, 52.2% agreed).	Positive
Kong and Yang (2024)	Pre-experimental	Paired sample t tests	Cognitive Engagement	Higher-order Thinking: Students reported a significant enhancement in perceived reflection, with scores increasing from 3.64 (SD = 0.74) to 3.92 (SD = 0.66) ($M_{diff} = 0.28, p < 0.001$).	Positive Significant
Kong and Yang (2024)	Pre-experimental	Paired sample t tests	Cognitive Engagement	Concentration: Students reported an improvement in attention, with a mean increase from 3.59 (SD = 0.78) to 4.02 (SD = 0.68) ($M_{diff} = 0.43, p < .001$).	Positive Significant
Song et al. (2023)	Observational	Descriptive Statistics	Cognitive Engagement	Concentration: Students reported high engagement in concentration when using Virtual Go Mode ($M = 3.22, SD = 0.77$).	Positive
Emotional Engagement					
Zhexenova et al. (2020)	RCT	Mann-Whitney U tests	Emotional Engagement	Enjoyment: A statistically significant difference in Mood Change scores between the Robot (0.45 ± 0.68), Teacher (-0.05 ± 0.52), and Tablet (-0.05 ± 0.83) conditions ($U = 122, W = 312, Z = -2.35, p = 0.019$), indicating that the Robot condition was more enjoyable than the other two conditions.	Positive Significant

AI ON MOTIVATION AND ENGAGEMENT

Castellano et al. (2021)	Pre-experimental	ANOVA	Emotional Engagement	Enjoyment: Students reported positive emotional experiences interacting with robot ($F(4, 49) = 74.55, p < 0.030$). Students reported a significant increase in enjoyment from pre ($M = 2.38$) to post ($M = 2.83$) after interacting with Pepper ($F(2,48) = 10.12, p < 0.001$).	Positive Significant
Aslan et al. (2024)	Pre-experimental	Paired samples T-test	Emotional Engagement	Enjoyment: Students reported a significant positive change in their pre- and post-ratings of emotions ($t(13) = -3.31, p = .006, d = 0.88$).	Positive Significant
Mubin et al. (2019)	Quasi-experimental (post & control)	ANOVA	Emotional Engagement	Enjoyment: Children in the robot condition displayed significantly more positive facial expressions ($F(1,42) = 13.01, p = 0.001$), smiling for 9% of the session duration compared to 2% in the human condition.	Positive Significant
Cheng et al. (2024)	RCT	Paired samples T-test	Emotional Engagement	Enjoyment: More participants in the treatment group felt happy reading the letters than in the control group, with fewer disagreeing (44.0% vs. 56.5%) and more strongly agreeing (20.0% vs. 8.7%).	Positive
Qian et al. (2023)	RCT	ANCOVA	Emotional Engagement	Enjoyment: Children learning with IARE rated it higher than those using flashcards in Enjoyment (5: 85.7% vs. 28.6%; 4: 14.3% vs. 0%; 3: 0% vs. 42.8%; 2–1: 0% vs. 14.3%).	Positive
Jauhiainen and Garagorry Guerra (2024)	Observational	Independent samples t-test	Emotional Engagement	Enjoyment: Approximately 66.4% of pupils reported substantial or high enjoyment when using the provided learning materials and laptops.	Positive
Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Emotional Engagement	Enjoyment: Students reported the robots creating an enjoyable English-speaking experience (I-Bot: 78.57%, P-Bot: 78.57%). Students expressed enthusiasm for CoolE Bot's diverse topic coverage (I-Bot: 60.71%, P-Bot: 67.86%).	Positive

AI ON MOTIVATION AND ENGAGEMENT

Lee (2024)	Observational	Descriptive Statistics	Emotional Engagement	Enjoyment: Students reported enjoyment in learning through AI (M = 4.30 out of 5, SD=1.11).	Positive
Ferreira et al. (2020)	Observational	Descriptive Statistics	Emotional Engagement	Enjoyment: 91.4% of children reported liking robot and enjoying the interaction. 88.6% of children reported high enjoyment of the activity.	Positive
Wen et al. (2025)	Observational	Descriptive Statistics	Emotional Engagement	Enjoyment: Students reported high enjoyment using AI-powered vocabulary learning system (M=4.071 out of 5, SD=0.986).	Positive
Ward et al. (2013)	Observational	Descriptive Statistics	Emotional Engagement	Enjoyment: 62% of children reported enjoying working with the intelligent tutoring system.	Positive
Ward et al. (2013)	Observational	Descriptive Statistics	Emotional Engagement	Excitement: 53% of children reported being more excited about science after using the intelligent tutoring system.	Positive
Lee et al. (2023)	Quasi-experimental (pre/post & control)	MANOVA	Emotional Engagement	Satisfaction: The experimental group reported higher satisfaction with the AI chatbot (M = 4.20/5).	Positive
Aslan et al. (2024)	Pre-experimental	Paired samples T-test	Emotional Engagement	Satisfaction: Students reported satisfaction in 91.3% of the sessions for emotional engagement.	Positive
Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Emotional Engagement	Satisfaction: Students expressed appreciation for CoolE Bot's cartoonish and human-like characters (I-Bot: 78.57%, P-Bot: 75%).	Positive
Lee (2024)	Observational	Descriptive Statistics	Emotional Engagement	Satisfaction: Students reported high satisfaction with the AI system (M = 4.14 out of 5, SD=1.01).	Positive
Tai and Chen (2024)	Quasi-experimental (post & control)	Descriptive Statistics	Emotional Engagement	Frustration: Students expressed frustration with advanced vocabulary and lengthy responses (I-Bot: 25%, P-Bot: 17.86%).	Positive

AI ON MOTIVATION AND ENGAGEMENT

Table 8

Excerpt of Qualitative Findings on Engagement

Category	Findings
General Engagement	General Engagement - “From watching the free play time footages, we observed that students revealed high engagement in learning.” (Dong, Choi, et al., 2024)
Behavioral Engagement	Learning Volume - “Within the control group, word counts appeared to decrease as the questions went on, while they appeared fairly level in the treatment group.” (Cheng et al., 2024)
	Time - “Students were engaged with using AI Duet to create piano music and Microsoft Image Creator from Microsoft Designer to prompt images as they continued to use these AI tools even in their free-play time.” (Dong, Mitchell, et al., 2024)
	Participation Attempts - “P15 (control), for instance, wrote "I don't know" or an equivalent to every question. Another participant (P50, treatment) tested the system by responding with messages like "Hi" and "I don't know" but finally wrote an earnest and relevant answer after five reprompts from the LLM.” (Cheng et al., 2024)
Emotional Engagement	Enjoyment - “The most common feedback regarding the strengths was that the AI platform was perceived as "fun". The majority of the students expressed that while the traditional textbook was boring, the AI systems offered interesting and fun ways to study.” (Lee, 2024)
	Satisfaction - “When it recognized my speech, I felt so satisfied.” (Jeon, 2024)
	Excitement - “Children mentioned "exciting," "awesome," and "interesting" only in the post-program interview, which represents a higher level of positivity towards social robots after attending the program.” (Dong, Choi, et al., 2024)
	Frustration - “One I-Bot participant expressed frustration: The bot talks too much. Some sentences are too long. It is hard to understand.” (Tai & Chen, 2024)
	Boredom - “It can be boring sometimes. Because, it's like, you get one task right, and you might get that same task for 15 or 20 more minutes. The exact same kind of task. Before they understand that you know it. That you get it.” (Moltudal et al., 2020)

AI ON MOTIVATION AND ENGAGEMENT

Cognitive Engagement	Patience	- “Some children demonstrated patience and concentration while waiting for the robots to start any actions.” (Dong, Choi, et al., 2024)
	Concentration	- “Children were attentive and responsive whenever the robots followed their commands.” (Dong, Choi, et al., 2024)
	Higher-order Thinking	- “It helps me think about whether something is missing from my text-prompts.” (Chen et al., 2024)
Agentic Engagement	Question Asking	- “Children showed a strong desire for learning robotics by asking researchers higher level questions regarding social robots as the program proceeded.” (Dong, Choi, et al., 2024)
	Tool Application	- “As indicated in the interview data, students could employ other digital tools such as online dictionaries or search engines while using the chatbots, which helped sustain their willingness to interact with the chatbots using English.” (Jeon, 2024)
	Strategy Adjustment	- “The participants self-monitored and evaluated their interactions with the chatbot, adjusting their communicative strategies accordingly.” (Tai & Chen, 2024)